A REPORT

ON

DEVELOP A VCU BASED DIGITAL BATTERY PASSPORT FOR TRACKING BATTERY USAGE IN ACCORDANCE WITH EU ESPR 2027

BY

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BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

NOVEMBER 2026

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Prepared in partial fulfilment of the

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AT

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BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

NOVEMBER 2026

Certificate

This is to certify that the Dissertation entitled **"Develop a VCU Based Digital Battery Passport for Tracking Battery Usage in Accordance with EU ESPR 2027"** and submitted by **Mr. Nithyanandham S**, ID No. **2024HT65556**, in partial fulfilment of the requirements of **AELZG628T Dissertation**, embodies the work done by him under my supervision.

Signature of the Supervisor

Place: Chennai

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Date: _____

Designation: Deputy Manager, AVL Technical Centre Pvt. Ltd, Chennai

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ABSTRACT

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Title of the Project: Develop a VCU Based Digital Battery Passport for Tracking Battery Usage in Accordance with EU ESPR 2027

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Key Words: Digital Battery Passport; EU Battery Regulation 2023/1542; State of Health; Vehicle Control Unit; SHA-256 hash chain; ECDSA; append-only database; tamper evidence

Project Areas: Automotive Electronics; Systems Engineering; Battery Management Systems; Embedded and Cloud Software

Abstract:

Regulation (EU) 2023/1542 requires a Digital Battery Passport (DBP) for EV, light-means-of-transport and industrial batteries above 2 kWh from February 2027. Its dynamic data, such as State of Health (SoH), is produced by the manufacturer's own electronics, so the passport must be both available and trustworthy. This work designs, builds and verifies an open, end-to-end DBP system. A Simulink BMS model and STM32 firmware provide the passport signals, verified on the CAN bus at 500 kbit/s. A VCU computes the Annex VII metrics and sends records to a cloud service that stores them in an append-only database, links them in a SHA-256 hash chain and serves them through three access tiers and a public portal. An ESP8266 controller integrates BMS, VCU and gateway and adds vehicle-side trust: a fingerprint of the SoH algorithm in every record, an independent VCU capacity cross-check, ECDSA-signed records and an offline outbox. On the live system the portal verified 604 chained records of a test battery, an independent auditor tool verified 310 signed records with no gaps, and a deliberate +5 % SoH manipulation was detected (6.60 % deviation against a 3 % limit) and logged.

_____ _____

Signature of Student **Signature of Supervisor**

Date: _____ Date: _____

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List of Abbreviations

API	Application Programming Interface
BIN	Battery Identification Number
BMS	Battery Management System
CAN	Controller Area Network (ISO 11898)
CORS	Cross-Origin Resource Sharing
DBP	Digital Battery Passport
DoD	Depth of Discharge
ECDSA	Elliptic Curve Digital Signature Algorithm (here on curve P-256)
EFC	Equivalent Full Cycle
ESPR	Ecodesign for Sustainable Products Regulation
EU	European Union
HMAC	Hash-based Message Authentication Code
HTTPS	HTTP over Transport Layer Security
INCOSE	International Council on Systems Engineering
JSON	JavaScript Object Notation
LMT	Light Means of Transport
LUT	Look-Up Table
MCU	Microcontroller Unit
OCV	Open-Circuit Voltage
P-256	NIST elliptic curve secp256r1
RTC	Real-Time Clock
SE	Systems Engineering
SHA-256	Secure Hash Algorithm with a 256-bit output (FIPS 180-4)
SoC	State of Charge
SoCE	State of Certified Energy
SoH	State of Health
StRS / SyRS	Stakeholder / System Requirements Specification
TLS	Transport Layer Security
UDS	Unified Diagnostic Services (ISO 14229)
VCU	Vehicle Control Unit

Chapter 1 – Introduction

1.1 Background and Motivation

Traction batteries have become the defining component of the electrified vehicle — carrying the highest material cost, the greatest environmental impact per kilogram, and the most complex lifecycle trajectory of any automotive part. As electrification scales globally [1], the automotive industry faces a structural information problem: the entities responsible for servicing, repurposing, and recycling batteries at end-of-life rarely have access to the detailed performance history and composition data needed to make sound technical and economic decisions.

The absence of a standardised, machine-readable battery record creates friction across the entire value chain. Recyclers cannot efficiently sort packs without physical dismantling. Repurposers cannot assess second-life viability without expensive re-testing campaigns. Regulators cannot enforce extended producer responsibility without lifecycle traceability. The result is suboptimal material recovery, underutilised second-life potential, and growing regulatory exposure for manufacturers.

The European Union has responded, through Regulation (EU) 2023/1542 [2] and the Ecodesign for Sustainable Products Regulation [3], by mandating the Digital Battery Passport (DBP) as a condition of placing EV traction batteries on the EU market from February 2027. The DBP is a structured, machine-readable electronic record that must accompany each battery throughout its lifecycle — from manufacture through first use, any second-life deployment, and end-of-life processing. A passport is only useful if the people who rely on it can trust it: the dynamic part of the record (State of Health, cycle count, negative events) is produced inside the vehicle by the battery maker's own electronics, and its value to a buyer, insurer or second-life operator depends on being able to show that it has not been altered. This dissertation addresses both halves of the problem — producing the data and making it trustworthy — within the practical constraints of an automotive electronics laboratory.

1.2 Problem Statement

From February 2027 every EV, light-means-of-transport and industrial battery above 2 kWh placed on the EU market needs a passport whose dynamic part — State of Health, cycle count, negative events — is generated inside the vehicle. Three questions follow for a vehicle maker: where the passport data should be computed (BMS, VCU or cloud); how the record can be stored so that later changes are detectable by any reader; and how a reader can trust values that come from the manufacturer's own electronics, including in vehicles that are not permanently connected. This dissertation addresses these questions with a working prototype rather than a paper design.

1.3 Objectives

The objectives of the dissertation are:

1. Design and implement a BMS controller Simulink model that simulates battery behaviour and produces the seven required DBP output signals with documented accuracy.
2. Define a requirement specification covering BMS, VCU and DBP data management using INCOSE SE methodology.
3. Develop BMS firmware for an STM32F103C8T6 MCU and validate all CAN signal outputs with a TOSUN CAN SPY at ACC Chennai.
4. Provide a structured SE justification for computing battery lifecycle status in the cloud/VCU layer rather than on the BMS.
5. Analyse DBP use cases and data-integrity scenarios, including deliberate hardware replacement, and propose mitigations.
6. Implement the VCU and the cloud DBP store end to end — append-only records, SHA-256 hash chain, three-tier access API and a public portal — and verify them on live data.
7. Make the passport trustworthy beyond the cloud: make the SoH algorithm verifiable, allow vehicles without permanent connectivity to keep a trustworthy record, and keep the DBP logic independent of the controller it runs on.

1.4 Scope and Limitations

The work covers the full chain from battery signals to a public passport portal: a BMS model and firmware, two VCU implementations, a cloud passport service with three access tiers, and vehicle-side trust mechanisms. It was carried out with a simulated battery and test battery identifiers; one real battery (-001) holds only a static passport. The following items are outside the scope: real battery cells on the ESP8266 path (the battery is simulated), production-grade identity (OAuth 2.0 / X.509 / hardware key storage), and formal mapping to the EU DPP common data model, which is still awaiting its implementing act.

The main limitations of the results are that the ESP8266 battery data is simulated, that identity and key storage are at prototype level, and that free-tier hosting is used. They are listed in full in Section 12.2.

1.5 Methodology

The work followed the systems engineering V-model of the INCOSE handbook [4]. On the left side, stakeholder needs were taken from Regulation (EU) 2023/1542, its Annex XIII and VII and the published passport guidance, and were broken down into stakeholder, system, component, cloud-interface and trust requirements (Chapter 3). Design decisions with real alternatives were recorded with their criteria (Chapter 6). The system was then built bottom-up: Simulink model, STM32 firmware, cloud service, VCU simulator and the ESP8266 integrated controller.

On the right side, each level was verified with the method that suits it: CAN signals with a TOSUN CAN SPY at ACC Chennai, the cloud service and portal on the live deployment, and the trust mechanisms with a planned manipulation of the SoH algorithm on running hardware. The data presented in Chapter 11 was collected directly from the live system — read-only SQL queries on the database, the portal itself, the Serial Monitor of the ESP8266 and the published auditor tool run on exported records — and every figure and table states the date of collection.

1.6 Organisation of the Report

Chapter 2 summarises the regulation and the literature surveyed. Chapter 3 presents the architecture and requirements, Chapter 4 the BMS model and Chapter 5 the lifecycle status logic. Chapter 6 records the main

design decisions. Chapters 7 and 8 describe the cloud service and the two VCU implementations, and Chapter 9 the mechanisms that make vehicle data trustworthy. Chapter 10 analyses integrity scenarios, Chapter 11 presents the results and Chapter 12 the conclusions, limitations and recommendations. Appendices, references and a glossary follow.

Chapter 2 – Regulatory Background and Literature Review

2.1 Regulatory Framework – EU 2023/1542

Regulation (EU) 2023/1542 [2], published in the Official Journal of the European Union on 28 July 2023, is the principal legislative instrument establishing the Digital Battery Passport. It applies to industrial batteries above 2 kWh, EV traction batteries and batteries for light means of transport placed on the EU market from the respective compliance dates. Annex XIII lists the information the passport must contain — general battery and manufacturer information, carbon footprint, supply-chain due diligence, material composition, performance and durability parameters, and end-of-life information — and Annex VII lists the parameters for determining State of Health. These attributes are the dataset that this project's BMS, VCU and cloud architecture generates, validates and stores.

Table 2.1 – EU Battery Regulation key requirements and compliance deadlines.

Regulation	Article	Key Requirement	Deadline
EU 2023/1542	Art. 77	Battery passport mandatory for EV, LMT and industrial batteries > 2 kWh	Feb 2027
EU 2023/1542	Annex XIII	Passport attributes (chemistry, SoH, carbon footprint, supply chain, end-of-life)	Feb 2027
EU 2023/1542	Art. 14 / Annex VII	State of Health and expected lifetime parameters made available to the battery owner and interested parties	Aug 2024 onwards
EU 2023/1542	Art. 77 / Annex XIII (1)–(4)	Three access levels: public, persons with a legitimate interest, notified bodies / authorities / Commission	Feb 2027
ESPR 2027	Art. 10	Data integrity and immutability of passport records (implemented here as a SHA-256 append-only hash chain)	Q4 2027
IEC 62620	Clause 5	SoH thresholds for second-life qualification (> 80 % high-performance, > 70 % residential)	Ongoing
ISO 11898 / ISO 15765-2 [5]	—	CAN physical / data-link layer and transport layer for in-vehicle data	Ongoing

2.2 Three-Tier DBP Access Model

Regulation (EU) 2023/1542 separates passport data by who may see it. The architecture of this project supports three tiers:

- **Tier 1 — Public:** chemistry, manufacturer, model, rated capacity and voltage, carbon footprint, battery status, second-life potential and a QR code. Accessible to anyone without authentication.
- **Tier 2 — Interested persons (OEM, workshop, second-life operator):** all Tier 1 data plus the in-use dataset — SoH, SoCE, remaining capacity and power, capacity and power fade, internal resistance increase, round-trip efficiency, energy and capacity throughput, time at extreme temperature, deep-discharge, overcharge and accident counters.
- **Tier 3 — Authorities / notified bodies:** all Tier 2 data plus regulatory data (supply chain, certification reference, end-of-life plan, audit trail) and the full record history needed to verify the hash chain.

In the target design Tier 2 is reached through OAuth 2.0 and Tier 3 through an X.509 certificate. The prototype implements the tiers with server-verified accounts and signed, expiring session tokens (Section 7.5); the production authentication methods are listed as future work.

2.3 Digital Product Passport Standards

The concept of a Digital Product Passport emerged from EU sustainable product policy and has been studied across multiple industries. Morales et al. [6] demonstrated the use of the Asset Administration Shell (AAS) serialisation format for representing battery digital passports in a machine-readable, vendor-neutral format, aligned with the common data model expected from IEC 63386 (under development).

Berger et al. [7], in work carried out with AVL List GmbH, derived a UML information model for a battery passport from expert interviews with end-of-life and second-use actors. Their results show that recyclers mainly need beginning-of-life data (chemistry, disassembly information) while second-use operators need middle-of-life data such as battery status and maintenance history. This supports the emphasis of the present work on trustworthy in-use data (SoH, cycles, negative events). Gianvincenzi et al. [8] proposed a standardised, tree-structured battery passport data model that maps Regulation 2023/1542 attributes into a life-cycle-inventory-ready structure; the field grouping used in this project's API (static Annex XIII sections, in-use Annex VII parameters, regulatory data) follows the same separation.

The EU Digital Product Passport technical specification [9] defines the API and data exchange expectations for passport systems, including tiered access. Industry implementations by large OEMs are typically proprietary; the academic literature highlights the need for open approaches that small pack assemblers and integrators can adopt without large IT investment — the gap this project's open repository addresses.

2.4 CAN Bus in BMS Applications

The Controller Area Network (ISO 11898) [10] is the dominant physical and data-link layer for BMS communication: differential signalling is robust to powertrain noise, non-destructive bitwise arbitration gives deterministic access, and 500 kbit/s offers a good balance of timing margin and throughput. All BMS signals in this work fit standard 8-byte frames with 11-bit identifiers on the STM32F103C8T6 bxCAN peripheral, validated with the TOSUN CAN SPY at ACC Chennai.

2.5 Second-Life Battery Assessment

IEC 62620 [11] thresholds place batteries above 80 % SoH in high-performance second-life applications and those between 70 % and 80 % in lower-demand stationary storage. The economics of second life depend on the cost and reliability of assessing state at retirement; a trustworthy record of SoH history, cycle count and negative events reduces the need for full electrochemical re-testing. These thresholds set the accuracy requirement of the SoH signal and the lifecycle logic in Chapter 5.

2.6 Tamper Evidence: Hash Chains, Signatures and Blockchain

Kouhizadeh et al. [12] examined blockchain for immutable product records in circular-economy applications. A blockchain combines three ideas: hash-linking of records (Haber and Stornetta [13]), digital signatures of the writers, and distributed consensus among parties that do not trust each other. For a battery passport the first two provide almost all of the value — any change to a stored record breaks every later link, and a signature binds the data to the device that produced it — while consensus adds cost and complexity that a regulated, single-operator passport service does not need.

This project therefore uses a **SHA-256 hash chain** (FIPS 180-4 [14]) on the server and **ECDSA P-256 signatures** (FIPS 186-5 [15]) on the device, without a blockchain. Tamper evidence does not depend on trusting the database operator: anyone holding the records can recompute the chain, and anyone holding the device public key can verify the signatures. The trade-off is analysed as SE Decision 2 in Chapter 6.

Chapter 3 – System Architecture & Requirements

3.1 BMS / VCU / Cloud Layered Architecture

The architecture separates raw signal production (BMS), derived passport metrics and gateway functions (VCU), and storage, access control and verification (cloud). The split is motivated by separation of concerns (safety-critical BMS functions versus regulatory DBP functions), independent update paths for the DBP formula set, and the greater computing resources available above the BMS MCU.

Table 3.1 – BMS / VCU / cloud layer responsibilities as built.

Layer	Platform (as built)	Responsibility	Output
BMS controller	Simulink model + STM32F103C8T6; BMS module in ESP8266 firmware	Battery simulation, BIN, voltage, capacity, current, SoC, SoH, cycle count	CAN 500 kbit/s (STM32) or internal variables (ESP8266)
VCU	Raspberry Pi / Python VCU kit and simulator; VCU module in ESP8266 firmware	Annex VII metrics, event detection, SoH cross-check, record signing, store-and-forward	HTTPS POST /api/v1/dbp/record
Cloud DBP API	Python 3 / Starlette on Render (free tier)	Validation, canonical JSON, SHA-256 chain, tier enforcement, QR codes	REST JSON (Tier 1/2/3)
Cloud DBP store	Supabase PostgreSQL	battery_passport (static) and dbp_records (append-only, triggers)	PostgREST
Portal	Static HTML/JS on GitHub Pages	Tier views, live update, full-chain verification in the browser	Web UI

3.2 Functional Block Diagram

Figure 3.1 shows the functional block diagram of the BMS → CAN → VCU → cloud chain. Figure 3.2 shows the complete system as implemented. Two vehicle-side paths feed the same cloud service: Path A follows the BMS → CAN → VCU split, and Path B integrates the battery model, BMS, VCU and gateway in a single ESP8266, which also carries the trust features described in Chapter 9.

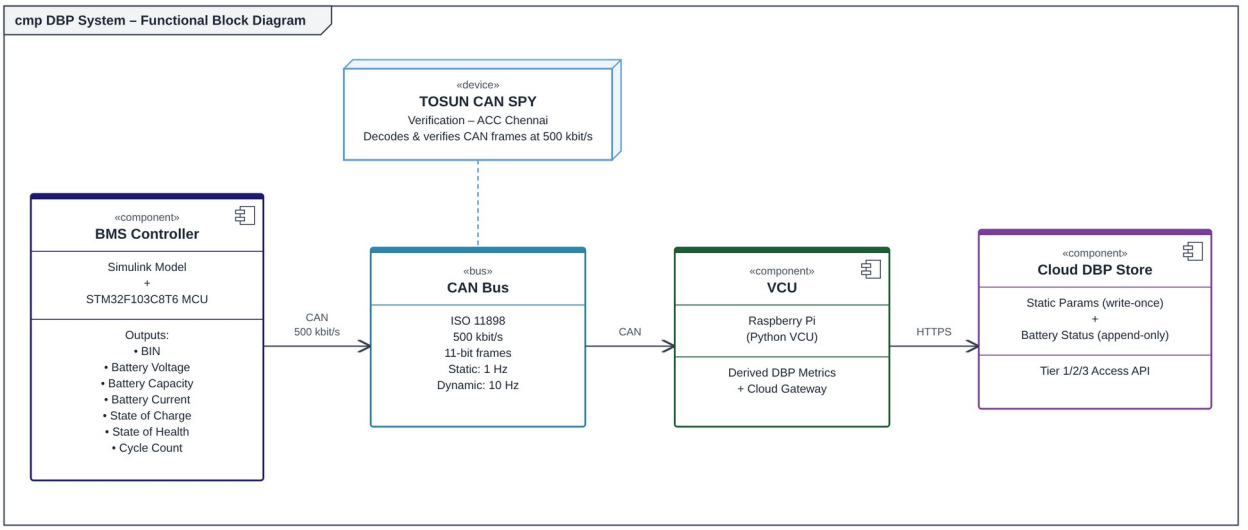


Figure 3.1 – Functional block diagram: BMS controller, CAN bus, VCU and cloud DBP store (UML component notation).

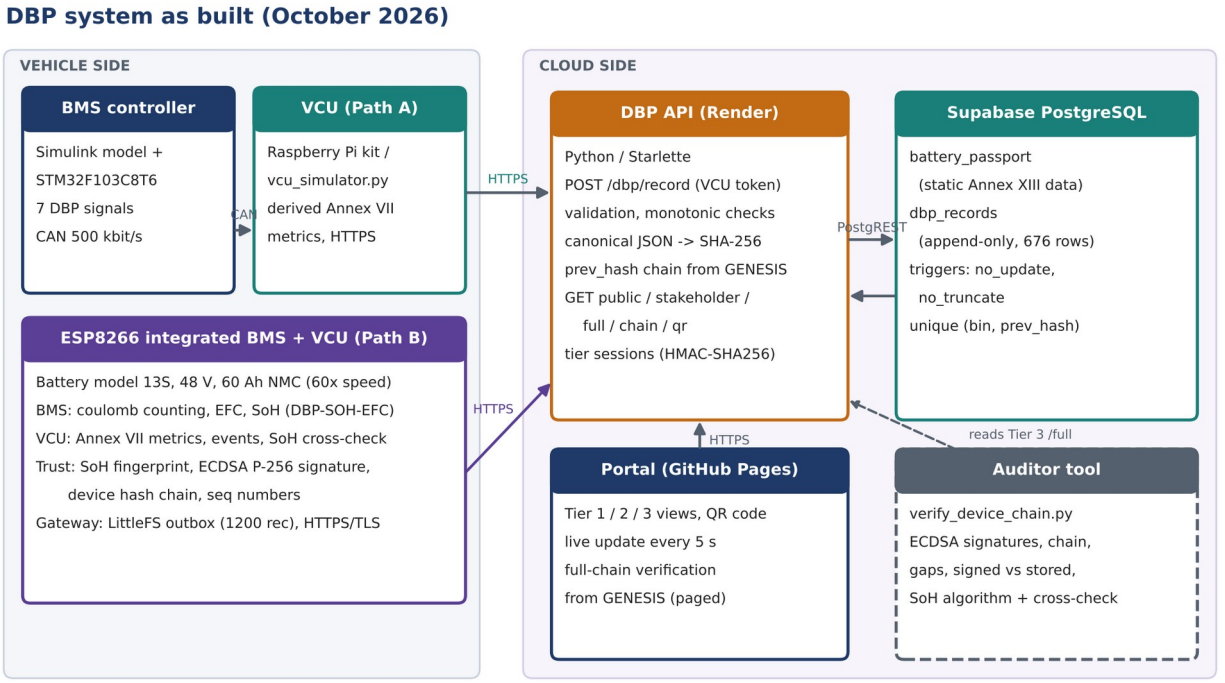


Figure 3.2 – DBP system as built: two vehicle-side paths, cloud API, append-only store, portal and auditor tool.

3.3 Stakeholder Interaction Flow

Figure 3.3 shows how each stakeholder interacts with the DBP across the battery lifecycle — from commissioning by the battery manufacturer through active service, workshop maintenance and second-life repurposing — and which tier each uses.

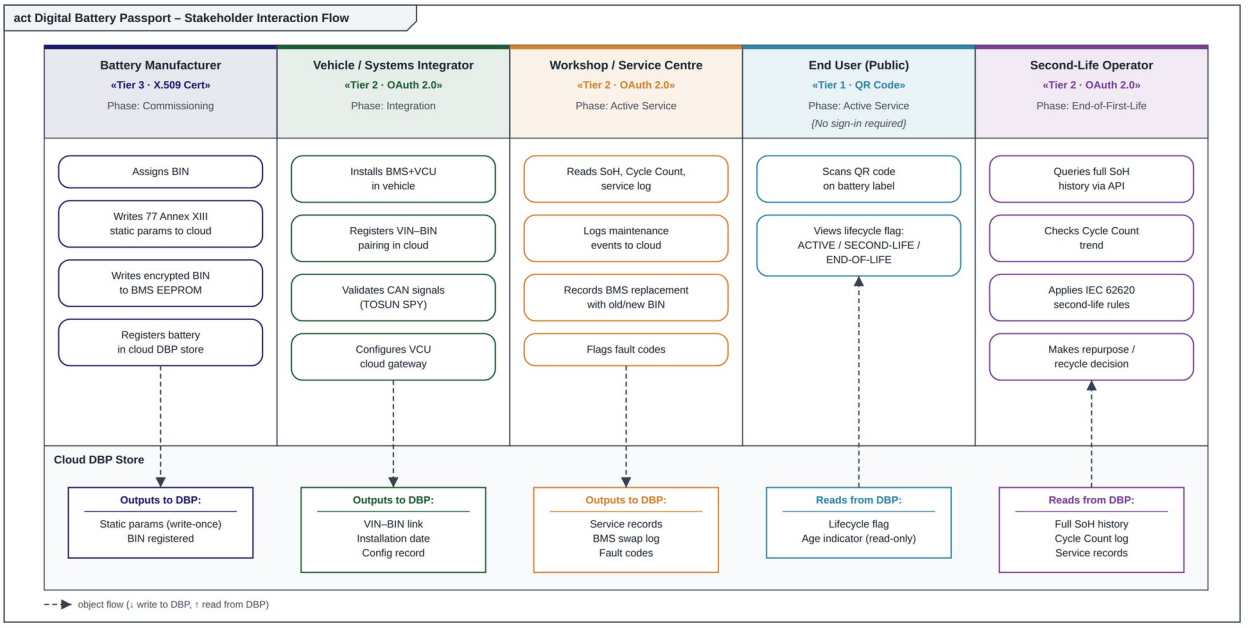


Figure 3.3 – Stakeholder interaction flow: five actor types, access tier, actions and data exchanged with the DBP store.

3.4 System Requirements Specification

Requirements follow a four-level hierarchy per INCOSE SE Handbook v4 [4]: Stakeholder Requirements (StRS), System Requirements (SyRS), Component Requirements (CRS) and Cloud Interface Requirements (SyRS-CLD). A further group, Trust Requirements (SyRS-T), covers the verifiability of the data produced on the vehicle.

3.4.1 Stakeholder Requirements (StRS)

Table 3.2 – Stakeholder Requirements (StRS).

ID	Requirement	Source
StRS-01	The DBP system shall produce the mandatory Annex XIII attributes for traction batteries.	EU 2023/1542 Annex XIII
StRS-02	DBP data shall be accessible to the public, verified stakeholders and authorities via a three-tier access model.	EU 2023/1542 Art. 77
StRS-03	The SoH signal shall have a measurement accuracy of ±1.5 % to support second-life qualification.	IEC 62620 / SE analysis
StRS-04	The SoC signal shall have a maximum error of ±2 % over a full charge/discharge cycle under nominal conditions.	SE analysis
StRS-05	All DBP data records shall be immutable once committed to the cloud store (append-only with hash chain).	ESPR 2027 Art. 10
StRS-06	The BMS shall transmit all seven DBP signals over CAN at 500 kbit/s within defined timing.	ISO 11898
StRS-07	The system shall detect and flag BMS replacement events to maintain DBP continuity.	EU 2023/1542 / SE decision

3.4.2 System Requirements – Static DBP Parameters (SyRS-S)

Table 3.3 – System Requirements – Static DBP Parameters (SyRS-S).

ID	Requirement	Verification method
SyRS-S-01	The BMS shall output a Battery Identification Number (BIN) as a fixed string at 1 Hz.	Inspection / CAN capture
SyRS-S-02	The BMS shall output the rated battery voltage with 0.1 V resolution.	Test / CAN decode
SyRS-S-03	The BMS shall output the rated battery capacity with 0.1 Ah resolution.	Test / CAN decode
SyRS-S-04	Static parameters shall be stored write-once in the cloud DBP store after commissioning.	Design review / cloud audit
SyRS-S-05	Static parameters shall be transmitted at 1 Hz on the CAN bus.	Test / TOSUN CAN SPY
SyRS-S-06	The BIN shall conform to EPCIS 2.0 format for supply chain traceability.	Design review

3.4.3 System Requirements – Dynamic Metrics (SyRS-D)

Table 3.4 – System Requirements – Dynamic Metrics (SyRS-D).

ID	Requirement	Verification method
SyRS-D-01	The BMS shall output SoC in the range 0–100 % with 0.5 %/LSB resolution.	Test / simulation
SyRS-D-02	SoC shall be computed by Coulomb counting with rated capacity as the denominator.	Model inspection
SyRS-D-03	SoC error shall not exceed ±2 % over a complete charge/discharge cycle.	Simulation test
SyRS-D-04	The BMS shall output SoH in the range 0–100 % with 0.5 %/LSB resolution.	Test / simulation
SyRS-D-05	SoH shall be computed by the Ah-throughput method referenced to the degradation curve.	Model inspection
SyRS-D-06	SoH accuracy shall be ±1.5 % or better.	Simulation / analysis
SyRS-D-07	The BMS shall output battery current in the range ±30 A with 0.01 A resolution.	Test / simulation
SyRS-D-08	Cycle count shall be a 16-bit unsigned integer, incremented once per complete cycle.	Test / simulation
SyRS-D-09	Dynamic signals shall be transmitted at 10 Hz on the CAN bus.	Test / TOSUN CAN SPY
SyRS-D-10	The inter-frame interval of dynamic CAN messages shall be 100 ms ± 2 ms.	Test / TOSUN CAN SPY

3.4.4 System Requirements – Data Integrity (SyRS-I)

Table 3.5 – System Requirements – Data Integrity (SyRS-I).

ID	Requirement	Verification method
SyRS-I-01	Every CAN frame transmitted by the BMS shall include a valid CRC per ISO 11898.	Test / CAN SPY error count
SyRS-I-02	The BMS firmware shall detect and flag CAN bus-off conditions within 100 ms.	Fault injection test
SyRS-I-03	Cloud DBP records shall be stored append-only with a SHA-256 hash of each record at write time.	Database audit / portal check
SyRS-I-04	The cloud DBP API shall enforce three-tier access control, rejecting unauthorised requests (HTTP 401/403).	API test

3.4.5 Trust Requirements (SyRS-T)

Table 3.6 – Trust Requirements (SyRS-T).

ID	Requirement	Verification method
SyRS-T-01	Every SoH value shall name the algorithm and calibration that produced it by a SHA-256 fingerprint; a change of fingerprint shall be logged as a chained event.	Inspection of records / events
SyRS-T-02	The VCU shall measure capacity independently over each full cycle and flag a deviation from the BMS SoH greater than 3 %.	Live test + manipulation demo
SyRS-T-03	Each record shall be signed on the device (ECDSA P-256) and chained to the previous device record with a sequence number.	Signature verification
SyRS-T-04	The vehicle shall buffer at least 12 h of records while offline and upload them in order when connected.	Design review / test
SyRS-T-05	An auditor shall be able to verify device signatures, chain, gaps and stored-versus-signed values without trusting the server.	Auditor tool test
SyRS-T-06	The portal shall verify the complete server hash chain from GENESIS, not only the most recent records.	Portal test on the full record history

3.5 Hardware and Software Platform Selection

Table 3.7 – Hardware and software platform selection.

Platform	Role	Key specifications
STM32F103C8T6 MCU	BMS controller hardware (Path A)	ARM Cortex-M3 @ 72 MHz, bxCAN, 12-bit ADC, 64 KB Flash, 20 KB SRAM
TOSUN CAN SPY	CAN bus verification	USB-CAN analyser, 500 kbit/s, DBC import, physical-value decoding (ACC Chennai)
Raspberry Pi + Python	VCU (Path A)	Linux, CAN via MCP2515 HAT, Python VCU kit; vcu_simulator.py (standard library only)
ESP8266 NodeMCU	Integrated BMS + VCU + gateway (Path B)	Tensilica L106 @ 160 MHz, 80 KB RAM, 4 MB flash (LittleFS), Wi-Fi, BearSSL TLS/ECDSA; Arduino IDE 2.3, esp8266 core 3.1.2
Render web service	DBP API hosting	Python 3, Starlette/uvicorn, manual deploy from GitHub; free tier sleeps after inactivity (~50 s cold start)
Supabase	DBP store	PostgreSQL with PostgREST; free tier pauses after about 7 days without activity
GitHub Pages	Public portal hosting	Static HTML/JS from docs/ of snithyamb/dbp-cloud

Chapter 4 – BMS Controller Model & Simulink Implementation

4.1 Model Overview and Design Philosophy

The BMS controller is implemented as a MATLAB Simulink model built from standard blocks and subsystems, with all parameters defined as workspace variables. This gives version-control-friendly sources, reproducible builds, parameterisation without structural change, and headless batch execution for regression testing. The model both simulates the battery electrical behaviour and implements the DBP signal algorithms that are replicated in the STM32 firmware, so it serves as verification reference and development prototype.

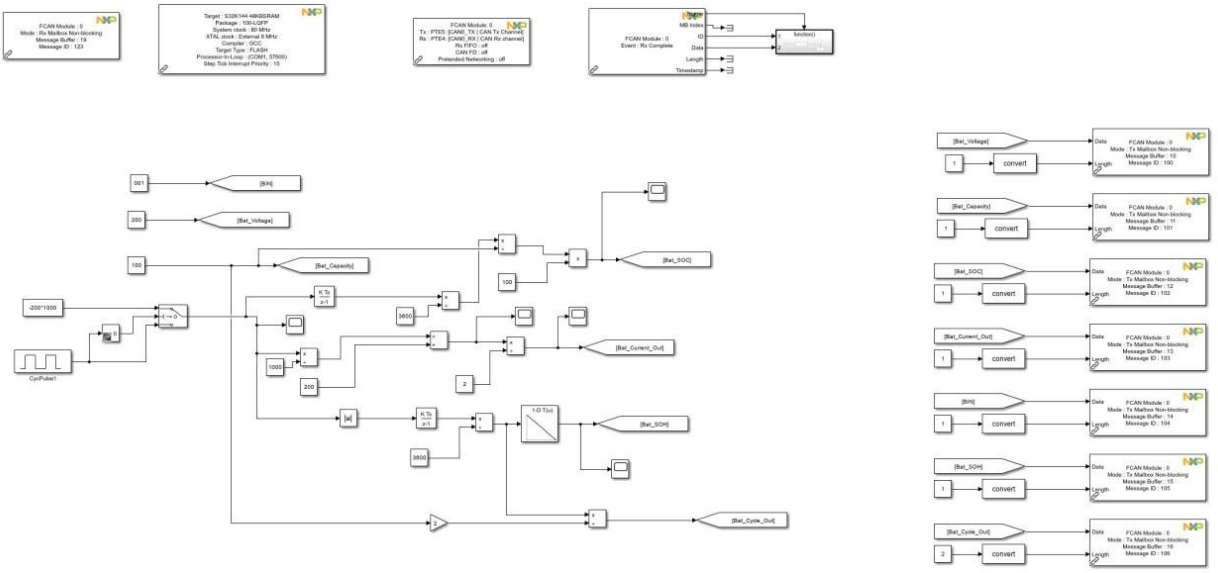


Figure 4.1 – BMS controller Simulink model producing the DBP output signals.

4.2 Battery Simulation Stimulus

The primary input is a Pulse Generator producing a symmetric charge/discharge current profile (amplitude 200, 50 % pulse width, period set to two full-cycle durations, sample time Ts = 0.01 s, i.e. 10× the CAN output rate). A switch separates discharge and charge half-cycles: the absolute value feeds Ah-throughput integration (SoH) and the signed value feeds Coulomb counting (SoC).

4.3 Signal Computation Blocks

Table 4.1 – BMS Simulink block description and computation logic.

Output signal	Simulink block(s)	Computation logic	Type / rate
Battery Identification Number	Constant (BIN_const)	Fixed alphanumeric battery identity	Static / 1 Hz
Battery Voltage	Constant (VBat_rated)	Rated pack voltage	Static / 1 Hz
Battery Capacity	Constant (CBat_rated)	Rated capacity at beginning of life	Static / 1 Hz
Battery Current	PWM generator → Gain	Scaled charge/discharge profile	Dynamic / 10 Hz
State of Charge	Integrator → Gain → Saturation	Coulomb counting normalised by rated capacity	Dynamic / 10 Hz
State of Health	Integrator → LUT → Saturation	Ah throughput mapped through degradation LUT	Dynamic / 10 Hz
Cycle Count	Edge detector → counter	Rising edges of the current profile = complete cycles	Dynamic / 10 Hz

4.4 SoC Computation – Coulomb Counting

Coulomb counting integrates battery current over time; the result is normalised by rated capacity and limited to 0–100 % by a saturation block, with the integrator initial condition set to the starting SoC. It is simple and accurate over short horizons, but current-sensor offset accumulates as drift. Model-based estimators that combine an equivalent circuit model [16] with Kalman filtering [17] limit this drift at a higher computational cost. The Simulink model uses an ideal source; the hardware uses the 12-bit ADC with careful scaling. Results (Chapter 11) show the SoC error inside ±2 % over two complete cycles.

4.5 SoH Computation – Ah-Throughput LUT Method

SoH is the remaining usable capacity relative to rated capacity. Capacity is lost mainly through loss of lithium inventory and loss of active material. The Ah-throughput method integrates |I| to obtain cumulative throughput and maps it through an empirically characterised 1-D look-up table to SoH; different chemistries are supported by replacing the LUT data. The method ignores temperature-accelerated ageing; a 2-D (throughput × temperature) LUT is the planned improvement.

In the ESP8266 integrated controller (Chapter 8) the same idea is expressed as a linear function of equivalent full cycles, `SoH = 100 − k_fade · EFC` with `EFC = Σ|I|Δt / (2·C_rated)` — the published, fingerprinted algorithm DBP-SOH-EFC v1.0.0 (Chapter 9).

4.6 Cycle Count and CAN Frame Encoding

Each complete charge/discharge cycle produces one rising edge, counted into a 16-bit unsigned counter (0–65 535). On the STM32 the count is derived from current-direction transitions with hysteresis and stored in Flash-emulated EEPROM. Two CAN frames are used: a static frame (BIN, voltage, capacity) at 1 Hz and a dynamic frame (current, SoC, SoH, cycle count) at 10 Hz, each signal scaled and offset into the available bit width and decoded by the TOSUN CAN SPY using matching signal definitions.

4.7 VCU Model

A companion Simulink VCU model (Figure 4.2) computes the derived DBP metrics that require vehicle-level integration: energy and capacity throughput, power and capacity fade, SoCE, time at extreme temperature and the deep-discharge, overcharge and accident counters. Its logic was carried into the Python VCU and the ESP8266 VCU module described in Chapter 8.

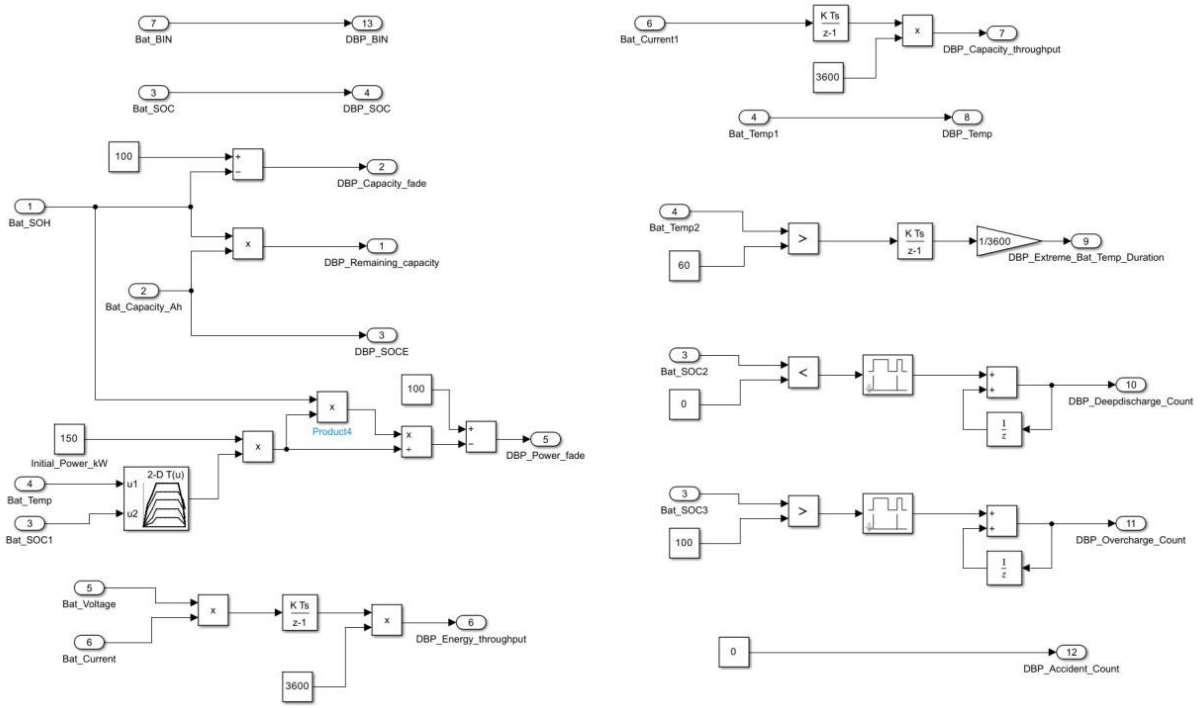


Figure 4.2 – VCU Simulink model computing the derived Annex VII metrics.

Chapter 5 – Battery Lifecycle Status

Battery lifecycle status indicates where a battery stands in its operational life — active traction service, approaching end of traction service, eligible for second life, or due for material recovery. It is distinct from real-time state: SoC and current answer "what is the battery doing now?", lifecycle status answers "how much useful life is left and what is the right next use?". The two primary inputs produced by the BMS are SoH and cycle count.

Table 5.1 – Battery lifecycle status: inputs, logic and outcome.

Lifecycle signal	Input	Logic	Lifecycle outcome
State of Health	|current| from ADC or model	Integrate |I| → Ah throughput → degradation LUT → SoH %	≥ 80 %: active traction; 70–80 %: second-life eligible; < 70 %: end of life
Cycle count	Current direction	Direction state machine with hysteresis → 16-bit counter → non-volatile storage	Complementary usage indicator; high cycles with high SoH flags possible manipulation
Lifecycle stage (cloud)	SoH history + cycle trend	Aggregate SoH over time, compute % loss per 100 cycles, apply IEC 62620 thresholds	ACTIVE / SECOND-LIFE ELIGIBLE / END-OF-LIFE on the public record

5.1 State of Health

SoH uses the absolute battery current, since every ampere in either direction contributes to ageing. The chain is: absolute current → integration per sample period → conversion to Ah → LUT mapping to SoH % → saturation to 0–100 % and CAN encoding at 0.5 %/LSB. In the cloud, the SoH time series determines the stage: ≥ 80 % ACTIVE, 70–80 % SECOND-LIFE ELIGIBLE (IEC 62620 stationary storage), < 70 % END-OF-LIFE.

5.2 Cycle Count

The cycle counter uses the sign of the current only. A two-state machine (CHARGING / DISCHARGING) with a hysteresis band around zero detects transitions; two confirmed half-cycles make one full cycle. The counter is persisted with timer-deferred writes to protect flash endurance. A high cycle count with low SoH is physically consistent; a high count with unusually high SoH is a manipulation indicator raised by cloud-side checks (Chapter 10).

5.3 Lifecycle Stage Determination

The stage flag is not computed by the BMS. The BMS produces raw SoH and cycle count, the VCU forwards them, and the cloud applies threshold comparison, trend analysis (rolling % SoH loss per 100 cycles), consistency cross-checks and publication on the Tier 1 record.

Implementation note: the cloud stores the Annex XIII 4(c) battery status (ORIGINAL, REPURPOSED, REMANUFACTURED, WASTE); the portal derives ACTIVE or SECOND-LIFE ELIGIBLE from it, and the server refuses any status change once a battery is WASTE. Automatic staging from the SoH trend is listed as future work.

Chapter 6 – SE Decisions

This chapter records the three main architectural decisions as trade studies following the INCOSE SE Handbook v4.

6.1 SE Decision 1 – Where to Compute Battery Lifecycle Status

The BMS (STM32F103C8T6) produces SoH and cycle count at 10 Hz, but lifecycle status needs history, standard thresholds, fleet context and remote access. Option A computes it on the BMS; Option B forwards raw values and computes it in the cloud.

Table 6.1 – SE Decision 1: on-BMS versus cloud lifecycle status computation.

SE concern	Option A: on BMS	Option B: in cloud	Driver
Computational resources	20 KB SRAM cannot hold a lifetime SoH series; flash endurance limits writes	Unlimited storage, full history queryable	B
Historical data	BMS knows only the current value	Every reading since commissioning is kept	B
Fleet normalisation	No fleet context	Compare against fleet baseline, detect outliers	B
Stakeholder access	Locked in EEPROM, needs physical CAN access	Published on Tier 1, reachable by QR code	B (Art. 77)
Integrity	Overwritten by firmware or BMS replacement	Append-only, independent of BMS hardware	B
Complexity	Lower at first; logic hard to update	Needs cloud infrastructure; logic updatable without re-flashing	A short term

Decision: Option B. The STM32 cannot keep a lifetime SoH history, the regulation requires remote multi-stakeholder access, and immutability cannot be guaranteed in on-board flash. This decision was implemented as described in Chapter 7.

Table 6.2 – Cloud interface requirements (SyRS-CLD).

ID	Requirement	Priority
SyRS-CLD-01	The cloud store shall receive SoH and cycle count from the VCU at each upload interval and store each record with a UTC timestamp.	High
SyRS-CLD-02	The cloud shall compute and update the lifecycle stage flag after each incoming SoH record, applying IEC 62620 thresholds.	High
SyRS-CLD-03	The lifecycle flag shall be published on the Tier 1 public record, accessible by QR code without authentication.	High

ID	Requirement	Priority
SyRS-CLD-04	Static battery parameters shall be stored in a write-once record at commissioning, locked thereafter.	High
SyRS-CLD-05	All stored records shall be append-only with a SHA-256 hash of each record.	Medium
SyRS-CLD-06	The API shall enforce three-tier access control (Tier 1 none, Tier 2 OAuth 2.0, Tier 3 certificate).	High

6.2 SE Decision 2 – Hash Chain versus Blockchain

Table 6.3 – SE Decision 2: integrity mechanism.

Criterion	Public / consortium blockchain	Hash chain + signatures (selected)
Tamper evidence	Yes	Yes — any change breaks all later links; recomputable by anyone holding the data
Origin of data	Writer signatures	Device ECDSA signature per record
Trust in operator	Not needed (consensus)	Not needed for detection; operator could withhold records, which sequence numbers expose
Cost / latency	Transaction fees or consortium operation; seconds to minutes	One PostgreSQL insert; milliseconds
Privacy (Tier 2/3 data)	Replicated ledger is hard to restrict	Normal database access control per tier
Fit for a regulated single-operator passport	Over-engineered	Adequate; can later anchor chain tips in a public ledger

Decision: a per-battery SHA-256 hash chain in an append-only table, plus device signatures. Periodically anchoring the latest chain hash in a public ledger or timestamping service remains possible without changing the design.

6.3 SE Decision 3 – Cloud Platform

The candidates were the large public clouds (AWS, GCP, Azure), a self-hosted server, and managed free-tier services. The selection criteria were zero cost for a student project, a managed relational database with triggers, HTTPS by default, deployment from a public Git repository, and the ability of the supervisor and examiners to open the portal without accounts. Render (API), Supabase (PostgreSQL) and GitHub Pages (portal) meet all of these. The accepted drawbacks are the Render cold start after inactivity and the Supabase pause after a week without activity; the API returns a clear HTTP 503 with CORS headers while the database is paused (Section 7.7).

Chapter 7 – Cloud DBP Implementation

This chapter describes the cloud side as deployed at https://dbp-cloud.onrender.com (API) and https://snithyamb.github.io/dbp-cloud/ (portal). Source code is in the public repository snithyamb/dbp-cloud.

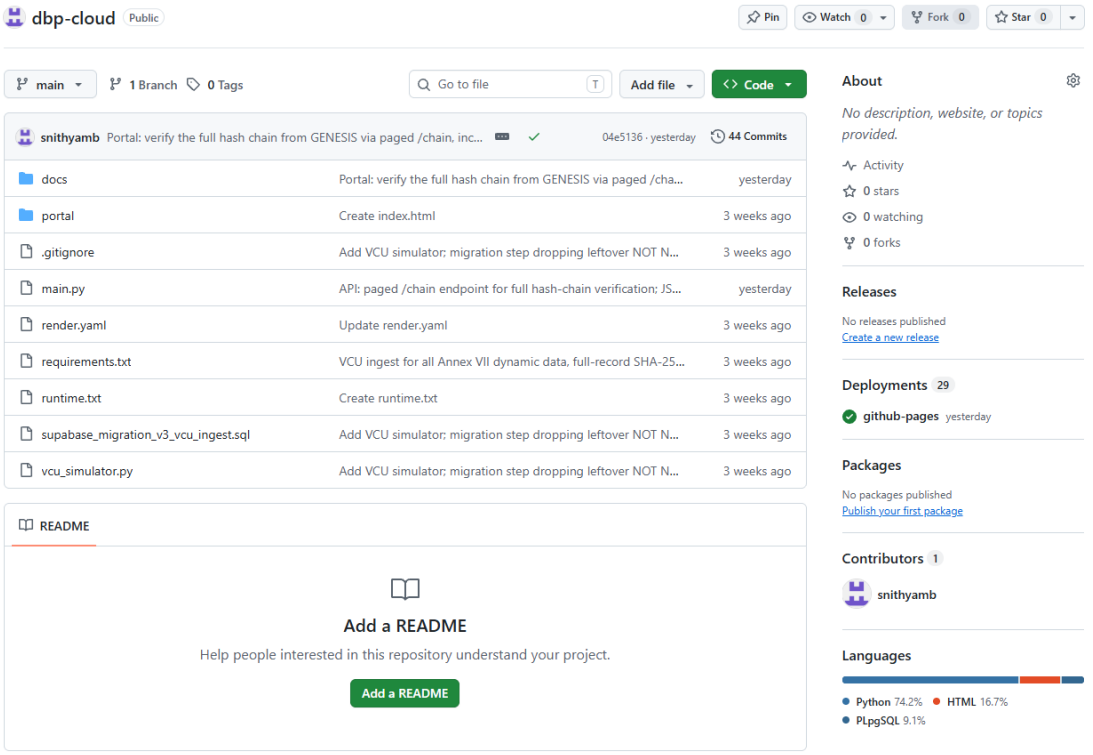


Figure 7.1 – Public repository snithyamb/dbp-cloud (44 commits; API, portal, VCU simulator and database migration).

7.1 Deployment

The API is a Python Starlette application served by uvicorn on a Render web service [18] connected to the main branch of the repository. Deploys are triggered manually from the repository. The health endpoint reports the active feature versions.

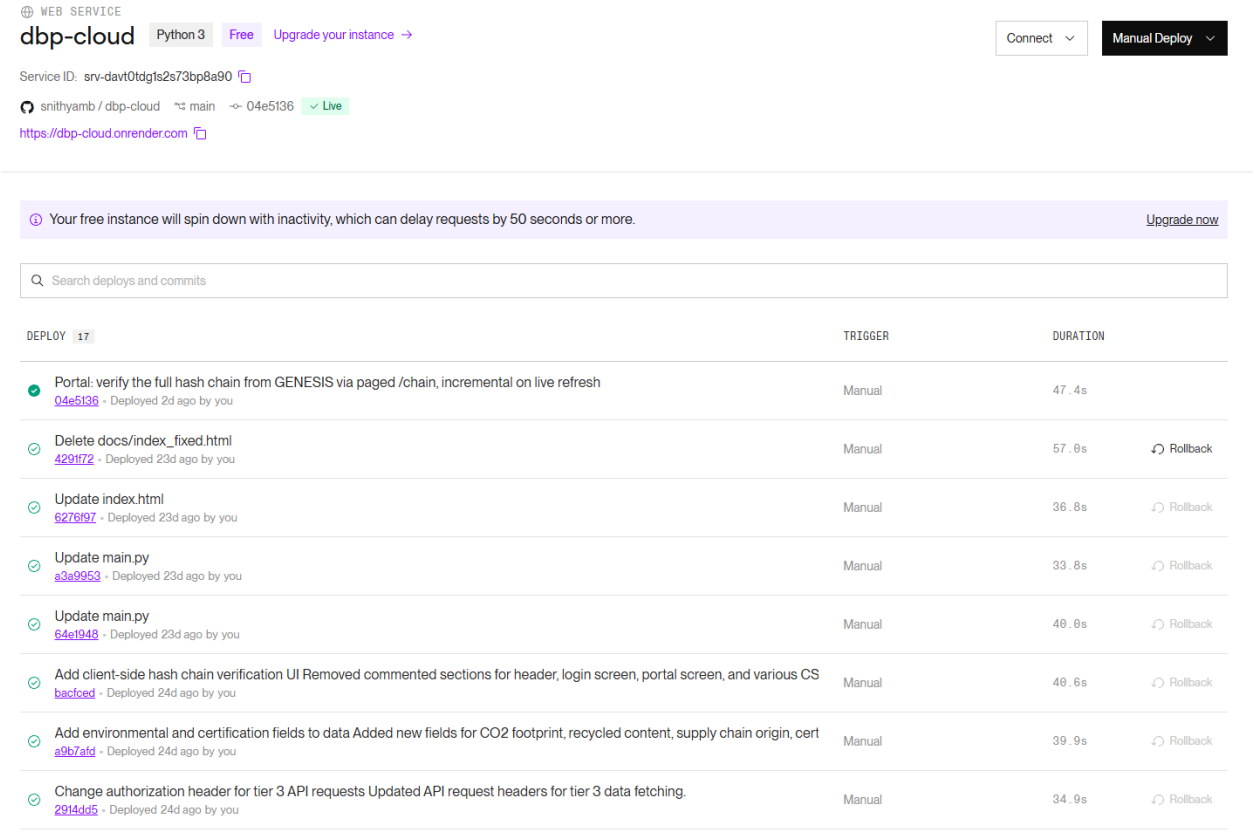


Figure 7.2 – Render dashboard: dbp-cloud web service (Python 3) connected to the GitHub repository.

```
GET https://dbp-cloud.onrender.com/health

{
  "status": "ok",
  "annex_xiii_fields": "v3",
  "vcu_ingest": "v2-full-hash",
  "chain_api": "v1-paged"
}
```

Figure 7.3 – Live /health response: status ok, Annex XIII field set v3, VCU ingest v2-full-hash, chain API v1-paged.

7.2 Data Model

Two tables hold the passport. **battery_passport** holds the static Annex XIII attributes per BIN (manufacturer, model, chemistry, rated values, carbon footprint, materials, regulatory references), written through the Tier 3 provisioning endpoint. **dbp_records** holds the dynamic record history: one row per TELEMETRY or EVENT record, with 25 numeric Annex VII / Annex XIII fields, the record type and event type, the device data (raw_telemetry), the canonical serialisation, prev_hash and record_hash.

Table 7.1 – dbp_records column groups.

Column group	Examples	Annex reference
State	soc_pct, soh_pct, soce_pct, remaining_capacity_ah, remaining_power_w	XIII 4(d), VII A
Fade / ageing	capacity_fade_pct, power_fade_pct, internal_resistance_increase_pct, rte_fade_pct, self_discharge_pct_per_month	IV A, VII A
Live measurements	pack_voltage_v, current_a, temp_celsius, depth_of_discharge_pct	XIII 4(d)
Cumulative (never decrease)	cycle_count, energy_throughput_kwh, capacity_throughput_ah, time_extreme_temp_h, deep_discharge_events, overcharge_events, accident_events	VII B
Status / events	lifecycle_status, record_type, event_type, event_detail, data_provider	XIII 4(c)
Integrity	canonical, prev_hash, record_hash, received_at, raw_telemetry.dev, raw_telemetry.soh_algo	Art. 10 (ESPR)

(a) id, bin, timestamp, SoC, SoH, cycle count

id int8	bin text	timestamp timestam...	soc_pct fl...	soh_pct fl...	cycle_count i...
6	DBP-2024HT65556-001	2026-10-03 18:26:39.7724	87.5	98.2	12
7	DBP-2024HT65556-002	2026-01-02 18:00:00+00	38.39	99.985	1
8	DBP-2024HT65556-002	2026-01-03 18:00:00+00	35.95	99.973	2
9	DBP-2024HT65556-002	2026-01-04 18:00:00+00	36.89	99.96	3
10	DBP-2024HT65556-002	2026-01-05 18:00:00+00	30.6	99.948	4
11	DBP-2024HT65556-002	2026-01-06 18:00:00+00	46.27	99.933	5
12	DBP-2024HT65556-002	2026-01-07 18:00:00+00	34.66	99.921	6
13	DBP-2024HT65556-002	2026-01-08 18:00:00+00	88	99.906	7
14	DBP-2024HT65556-002	2026-01-09 18:00:00+00	43.61	99.891	8
15	DBP-2024HT65556-002	2026-01-10 18:00:00+00	63.97	99.879	9
16	DBP-2024HT65556-002	2026-01-11 18:00:00+00	54.85	99.864	10

(b) the same rows: prev_hash → record_hash links and lifecycle status

id int8	prev_hash text	record_hash text	lifecycle_status text
6	GENESIS	NULL	IN_USE
7	GENESIS	7bb368778da7da3e77a1c4336274e73c	ORIGINAL
8	7bb368778da7da3e77a1c4336274e73c	b6a4ccbd277b69bb1fead9beadbc1cb6	ORIGINAL
9	b6a4ccbd277b69bb1fead9beadbc1cb6	a9d1326a6e75736043f6c91a3f52f41db	ORIGINAL
10	a9d1326a6e75736043f6c91a3f52f41db	40a6a55924cf7268532b0fe3ce6aa7e4	ORIGINAL
11	40a6a55924cf7268532b0fe3ce6aa7e4	014d7263e82fe18734468dbee60a92f4	ORIGINAL
12	014d7263e82fe18734468dbee60a92f4	d18bfeab045b65d7a140b66cde197f82	ORIGINAL
13	d18bfeab045b65d7a140b66cde197f82	16e677f0efe45e738b8b8a21ef87619cb	ORIGINAL
14	16e677f0efe45e738b8b8a21ef87619cb	32752026706380c56e614712af235b65	ORIGINAL
15	32752026706380c56e614712af235b65	11e0fd227eca2440acf2c8f748be19177c	ORIGINAL
16	11e0fd227eca2440acf2c8f748be19177c	961ae429506d55aaee3bc83928598b9	ORIGINAL

Figure 7.4 – Supabase table editor, dbp_records (excerpt of 769 records): (a) state columns, (b) prev_hash → record_hash chain links.

7.3 Record Ingest and Validation

A VCU sends POST `/api/v1/dbp/record` with a bearer token. Before anything is stored the server:

- rejects unknown fields and out-of-range values (for example SoC outside 0–100 %, temperature outside −60 to 150 °C);
- rejects a record whose cumulative counters (cycle count, throughput, event counts, time at extreme temperature) are lower than in the previous record — HTTP 409, which blocks a counter reset by a replaced or re-flashed BMS;
- rejects a timestamp earlier than the previous record and any status change after WASTE;
- accepts only the record and event types defined in the schema published at GET `/api/v1/dbp/schema`.

7.4 Canonical Serialisation and the Server Hash Chain

Every accepted record is serialised to one canonical string — keys sorted, no whitespace, numbers formatted exactly as JavaScript prints them, so that the browser can reproduce the same bytes — and the string includes `prev_hash`, the record_hash of the previous record of the same battery (or "GENESIS" for the first). Then `record_hash = SHA-256(canonical)`. The canonical text is stored verbatim, so any later verifier hashes exactly what was hashed at write time.

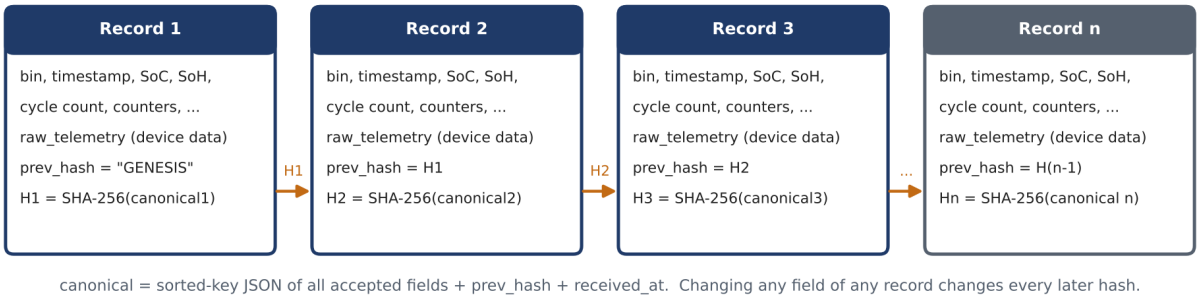


Figure 7.5 – Server-side SHA-256 hash chain per battery, starting from GENESIS.

Three database mechanisms make the history append-only: triggers `dbp_records_no_update` and `dbp_records_no_truncate` reject UPDATE, DELETE and TRUNCATE, and a unique constraint on (bin,

prev_hash) means two records can never claim the same predecessor — if two VCU uploads race, the second receives HTTP 409 and retries against the new chain head.

```
-- Records per battery and append-only protection (read-only)
select right(bin,3) as bin_suffix, count(*) as records,
       min(timestamp)::date as first_day, max(timestamp)::date as last_day,
       max(cycle_count) as max_cycles, min(soh_pct) as min_soh,
       count(*) filter (where raw_telemetry ? 'dev') as device_signed,
       (select string_agg(tgname, ', ') from pg_trigger
        where tgrelid = 'dbp_records'::regclass and not tgisinternal) as triggers
from dbp_records group by bin order by bin;
```

Results Chart

bin_suffix	records	first_day	last_day	max_cycles	min_soh	device_signed	triggers
001	1	2026-10-03	2026-10-03	12	98.2	0	dbp_records_no_truncate, dbp_records_no_update
002	164	2026-01-02	2026-05-13	132	98.215	0	dbp_records_no_truncate, dbp_records_no_update
003	604	2026-10-24	2026-10-31	252	95.63	312	dbp_records_no_truncate, dbp_records_no_update

Figure 7.6 – Read-only query on the live database: records per battery, device-signed records and the append-only triggers.

7.5 Access Tiers and Authentication

Tier 1 endpoints need no authentication. For Tier 2 and Tier 3 the portal sends the user's e-mail and password to `POST /api/v1/auth/login`; the server checks them against values held only in Render environment variables and returns a session token — a payload {sub, tier, exp} signed with HMAC-SHA256 and valid for one hour. Each request's tier is derived from that token (or, for machine clients such as the auditor tool, from a separate Tier 3 API key), never from anything the browser claims. Requests below the required tier receive HTTP 401/403.

Table 7.2 – DBP API endpoints.

Endpoint	Tier	Purpose
GET /health	—	Service status and feature versions
GET /api/v1/dbp/schema	—	Machine-readable description of what a VCU may send
POST /api/v1/dbp/record	VCU token	Append a TELEMETRY or EVENT record
POST /api/v1/auth/login	—	Issue a Tier 2/3 session token
GET /api/v1/dbp/{bin}/public	1	Public Annex XIII data, battery status, latest SoH
GET /api/v1/dbp/{bin}/stakeholder	2	In-use (Annex VII) dataset
GET /api/v1/dbp/{bin}/full	3	Regulatory data and the latest 50 records
GET /api/v1/dbp/{bin}/chain	3	Complete record history in id order, paged (≤ 1000 per page) for full-chain verification
GET /api/v1/dbp/{bin}/qr	1	QR code (SVG) linking to the public passport
POST /api/v1/dbp/provision	Tier 3 key	Create or update static battery_passport data
GET /api/v1/batteries	—	List of BINs with records

The portal labels Tier 2 as "OAuth 2.0" and Tier 3 as "X.509" to show the target design; in this prototype both are implemented with the server-verified login and HMAC-signed sessions described above.

7.6 Portal

The portal is a single static page on GitHub Pages. After login it shows the tier banner, a BIN lookup with optional live update every 5 seconds, headline SoC, SoH and cycle count, and then the data sections allowed for the tier: Tier 1 public data with the QR code, Tier 2 OEM / stakeholder data, Tier 3 regulatory data, the access-control summary and the immutability panel.

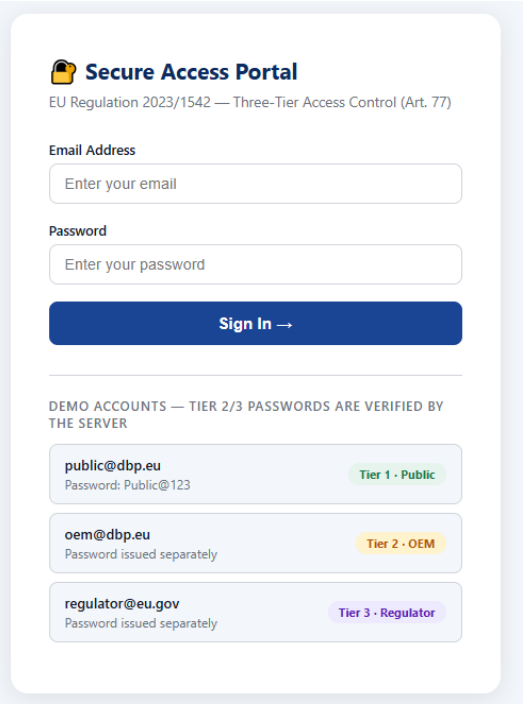


Figure 7.7 – Portal sign-in: three demonstration accounts, Tier 2/3 passwords verified by the server.

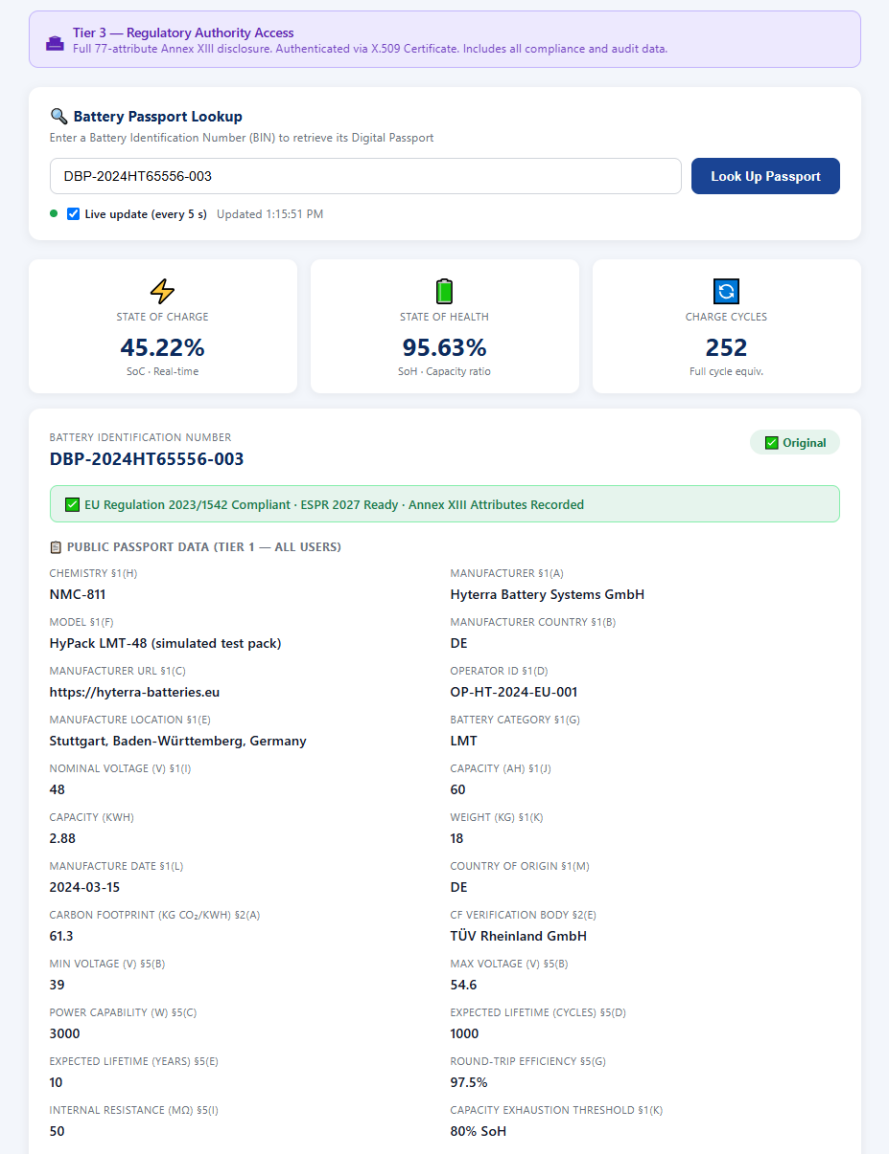


Figure 7.8 – Regulator (Tier 3) view of test battery DBP-2024HT65556-003: live SoC, SoH, cycles and Tier 1 data.



Figure 7.9 – Tier 1 QR code and Tier 2 in-use data reported by the ESP8266 (data provider ESP8266-BMS-VCU-01).

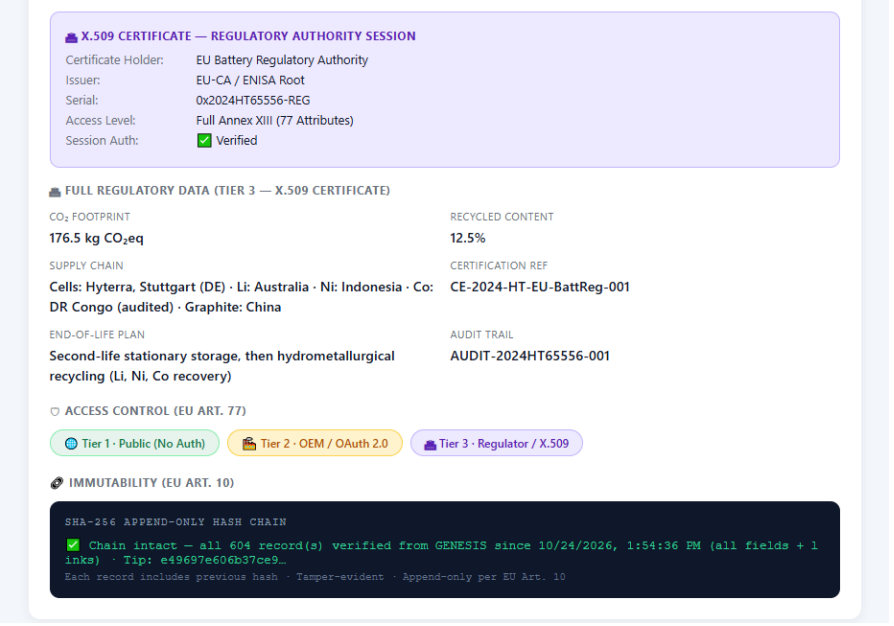


Figure 7.10 – Tier 3 regulatory data, access-control summary and the hash-chain verification result.

7.7 Full-Chain Verification in the Browser

A check of only the most recent records would miss a change to an older one. The portal therefore fetches the complete history through the paged `/chain` endpoint and checks every record from GENESIS: SHA-256 of the stored canonical text equals record_hash, and the canonical prev_hash equals the hash of the previous record. The result is cached; on live refresh only the new records are fetched and checked, so the check stays fast as the history grows.

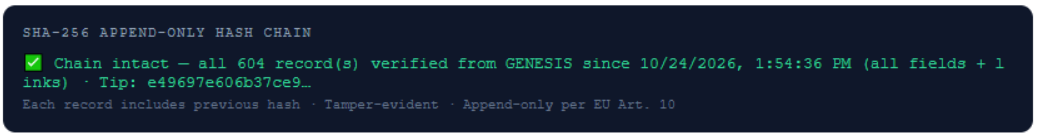


Figure 7.11 – Portal result on 31 Oct 2026: chain intact, all 604 records of -003 verified from GENESIS.

Each problem the check finds is named with its record id and kind. There are three kinds, matching the three ways a stored record can be altered: (1) **hash mismatch** — the canonical text was changed but record_hash was not recomputed; (2) **link to #N broken** — the attacker also recomputed record_hash of the changed record, so its hash is consistent again, but the next record still names the original hash in its prev_hash; (3) **stored value differs from hashed payload** — a database column was changed while the hashed canonical text was left alone. The portal then shows a red cross and "Tampering detected" with the list of problems instead of the green "Chain intact" result.

Figure 7.12 shows this on the live data of -003. Because the database rejects every UPDATE and DELETE on dbp_records, the alteration could not be made in the store itself; it was simulated in the regulator's browser by intercepting the `/chain` responses and changing the SoH of three records by +5 percentage points, one per kind: record #300 (99.65 → 104.65 %, canonical changed, hash kept), record #474 (99.38 → 104.38 %, canonical changed and record_hash recomputed) and record #600 (96.29 → 101.29 %, column changed only). All three were detected and named. When the interception was removed and the chain checked again, the portal returned to "Chain intact" for all records (Figure 7.13).

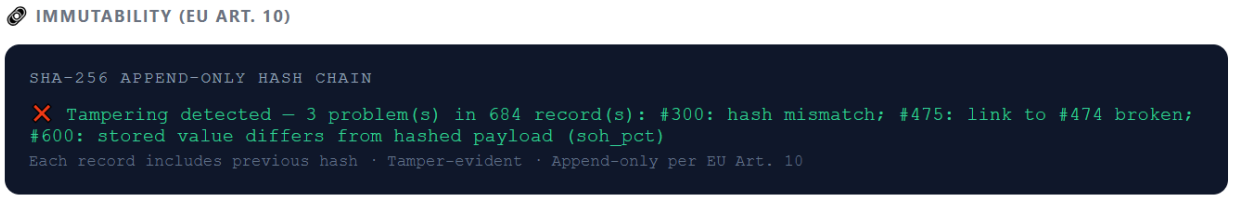


Figure 7.12 – Portal response to simulated tampering of three records of -003 (31 Oct 2026): hash mismatch at #300, broken link after the recomputed #474, and a stored SoH that differs from the hashed payload at #600.

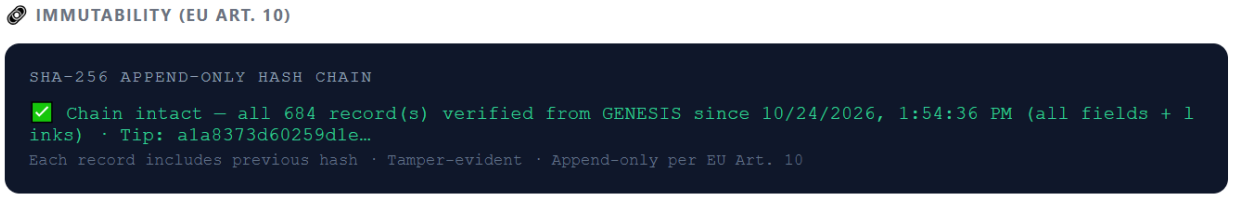


Figure 7.13 – The same check after the interception was removed: chain intact, all 684 records verified from GENESIS.

The API also handles a paused or restarting database: errors from the database client are converted into JSON responses with CORS headers (HTTP 503 "Database unavailable — it may be paused or restarting"), so the portal shows a meaningful message instead of a browser "Failed to fetch".

Chapter 8 – VCU Implementation

Two VCU implementations send data to the same API. Both use the same record schema, so the cloud cannot tell (and does not need to know) which kind of controller produced a record; the `data_provider` field records it.

8.1 Path A – Raspberry Pi VCU Kit and VCU Simulator

The VCU kit for the Raspberry Pi receives BMS CAN frames through an MCP2515 interface, computes the derived Annex VII metrics in Python and posts records over HTTPS. For repeatable cloud testing without the bench, `vcu_simulator.py` (Python standard library only) models a 120 Ah, 48 kWh pack with an empirical ageing model and pushes one TELEMETRY record per simulated day, plus EVENT records for deep-discharge, overcharge and thermal events. Test battery DBP-2024HT65556-002 holds a 132-day history created this way (164 records, January–May 2026 timeline, data provider VCU-SIM-01), including 29 thermal, 2 deep-discharge and 1 overcharge events.

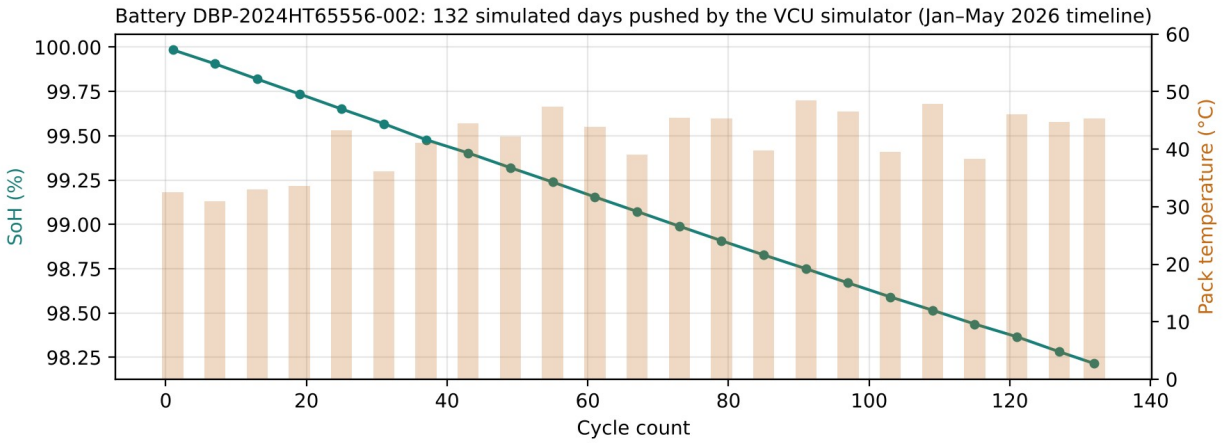


Figure 8.1 – SoH and pack temperature of test battery -002 over 132 simulated days (VCU simulator).

8.2 Path B – ESP8266 Integrated BMS + VCU + Gateway

To show that the passport functions do not need separate controllers or a CAN bus, a single Arduino sketch on an ESP8266 NodeMCU [19] runs four modules in one loop. The sketch refuses to compile for the real battery's BIN (-001), so the simulated battery can only ever write to test battery -003.

Table 8.1 – Modules of the ESP8266 integrated controller.

Module	Function	Key parameters
Battery model	13S NMC pack; OCV table 3.00–4.18 V per cell; R0 = 50 mΩ rising with ageing; I^2R heating with 20 kJ/K thermal mass and convective cooling; true ageing 0.021 % per EFC; 2 %/month self-discharge at rest	48 V, 60 Ah, 2.88 kWh, 3 kW; ambient 30 °C (Chennai)
Drive cycle	Urban-style discharge (0.3–0.6 C with pulses), rest 40 min, CC charge with taper, rest 60 min; every 5th cycle goes deep (4 % SoC)	SIM_SPEED = 60 (one cycle ≈ 7–8 real minutes)
BMS	Coulomb counting SoC; EFC counting; SoH by DBP-SOH-EFC v1.0.0 (k_fade = 0.020 %/EFC); fault flags for over/under-voltage and over-temperature	Persistent state with write-then-rename
VCU	Energy and capacity throughput, round-trip efficiency, SoCE, internal resistance from ΔV/ΔI, self-discharge during rest, DoD, time at extreme temperature, deep-discharge / overcharge / accident counters, rising-edge events, independent SoH cross-check	Extreme hot 45 °C, cold −10 °C; deep discharge < 5 % SoC; overcharge > 4.25 V/cell
Gateway	Builds, fingerprints and signs records; LittleFS outbox; HTTPS POST with certificate validation against ISRG Root X1/X2 and Google Trust Services R1/R4; NTP time	One TELEMETRY record per real minute; outbox 1200 records (≈ 20 h)

The Serial Monitor offers commands for demonstration: s prints status (including the SOH and SIGN lines), k prints the device public key, h toggles a +5 % SoH manipulation, and further commands inject overcharge, over-temperature and crash events to exercise the event path.

8.3 Store and Forward

Records wait in a flash outbox and are sent in order whenever Wi-Fi is available. The outbox holds 1200 records (about 20 hours at one record per minute). When the outbox is full, new telemetry is skipped but events are always kept. A record rejected by the server is moved aside so that it cannot block the queue, and counters and device state are written with write-then-rename so that a power cut cannot corrupt them.

8.4 Bring-up Issues

Table 8.2 – Issues found during integration and their fixes.

Symptom	Cause	Fix
Link error: ets_intr_lock undefined	Corrupted Arduino core build cache	Changed CPU frequency to 160 MHz to force a fresh core build
COM port busy / access denied	Second IDE window holding the port	Closed all IDE windows (File → Quit), re-plugged the board
Credentials overwritten by placeholders	Stale IDE window autosaved an old sketch	Wi-Fi and VCU token moved to a separate secrets.h that is never committed

Symptom	Cause	Fix
LoadStoreError at boot in br_ec_keygen	BearSSL key generation reads byte tables from flash in a way the ESP8266 cannot	Private key drawn from HMAC-DRBG; public key computed with the curve's mulgen() on the TLS helper stack

Chapter 9 – SoH Trust and Offline Integrity

An append-only cloud store with a hash chain proves that records were not changed after they arrived. It does not prove that the values were honest when they were produced, nor does it help a vehicle that cannot reach the cloud. Three problems remain, and they are most pressing for low-cost vehicles such as electric two-wheelers:

1. **Is the SoH algorithm honest?** The cloud stops anyone editing cycle count or old records, but an OEM could ship an SoH algorithm that reports optimistic values. How can the algorithm that produced each SoH value be identified and its output checked?
2. **No telematics.** In some markets a two-wheeler with a cellular module is regulated like a mobile phone, and many low-cost vehicles have no permanent connection at all. The passport must work offline and synchronise when possible.
3. **Where does the code run?** Some vehicles have no VCU — only a BMS or a telematics unit.

Table 9.1 – Summary of the trust mechanisms.

Problem	What was built	Status
1. SoH algorithm trust	SHA-256 fingerprint of the SoH algorithm and live calibration in every record; any change logged as a chained event	Running
1. SoH algorithm trust	Independent VCU capacity cross-check against the BMS SoH; event if $\|\Delta\| > 3$ %	Running; 261 PASS, 23 FAIL (demo)
1. SoH algorithm trust	Simulated manipulated BMS (+5 % SoH) to show both checks catch it	Demonstrated live, 31 Oct 2026
2. Offline operation	ECDSA P-256 signature and device hash chain on every record before upload	Running; 310 records verified
2. Offline operation	Flash outbox of about 20 h (1200 records)	Running
2. Offline operation	Workshop download over the diagnostic port (UDS)	Future work
3. Controller independence	All controllers combined in one ESP8266 build; portable core planned	Partly done
Auditor check	verify_device_chain.py: signatures, chain, gaps, stored vs signed values, SoH algorithm	Done

9.1 Making the SoH Algorithm Verifiable

9.1.1 SoH algorithm fingerprint

The BMS uses a defined, model-based algorithm, DBP-SOH-EFC version 1.0.0:

```
SoH_BMS = 100 − k_fade · EFC        EFC = Σ|I|·Δt / (2 · C_rated)
k_fade = 0.020 % per equivalent full cycle (80 % after 1000 cycles), C_rated = 60 Ah
```

At start-up the firmware hashes a text description of the algorithm and of the calibration it is actually running:

```
DBP-SOH-EFC|1.0.0|k_fade=0.0200|eol=80.0|rated_ah=60.0|offset=0.00
  → SHA-256 → 821963b754474cd356c27f634e0182fbf847e67c67daf7d7aaee2e9b91ae95d4
```

- The hash travels in every TELEMETRY record (`raw_telemetry.soh_algo`), so each SoH value names the algorithm that produced it.
- The first time a hash is seen, and whenever it changes, the device sends a `SOH_ALGORITHM` event with the full description and hash; the cloud chains it, so a later change of calibration cannot be hidden.
- An auditor holding the published algorithm and calibration can recompute the hash and confirm it.

9.1.2 Independent VCU cross-check

The VCU measures the real capacity over every full cycle from charge it counts itself and compares it with the BMS value:

```
SoH_VCU = ( Ah_discharged / (DoD / 100) ) / C_rated × 100        Δ = SoH_BMS − SoH_VCU
|Δ| ≤ 3 %  → PASS          |Δ| > 3 %  → FAIL + SOH_CROSSCHECK_FAIL event
```

- Only cycles of at least 30 % of rated Ah and at least 50 % depth of discharge are used; the first cycle after a reboot is skipped because it is only partly observed.
- Every telemetry record carries the result (PENDING / PASS / FAIL), the VCU-measured SoH and the deviation.
- In the simulation the cells age at 0.021 % per cycle while the BMS algorithm assumes 0.020 %, so the BMS reads slightly high, as a real algorithm would; the deviation stays inside the limit (Section 11.4).

9.1.3 Demonstrating a manipulated BMS

Typing `h` adds +5 % to the BMS SoH, simulating an OEM that inflates SoH. The calibration text changes (`offset=5.00`), so the fingerprint changes and a new `SOH_ALGORITHM` event is logged at once; at the next full cycle the VCU measures the true capacity, Δ ≈ +5 %, and logs `SOH_CROSSCHECK_FAIL`. An OEM could hide the change from the fingerprint only by misreporting its own calibration; it still cannot hide it from the cross-check, which uses current and SoC rather than the BMS SoH. The demonstration was carried out on the live system on 31 October 2026; the results are in Section 11.5.

Both checks are screening tools. In a dispute the reference remains a laboratory capacity test of the pack, for example to IEC 62660-1 [20] and the methods of Annex VII of Regulation (EU) 2023/1542.

9.2 Working Without Telematics

Records are made tamper-evident on the vehicle itself, so they can wait offline for hours or days and still be trusted when they reach the cloud.

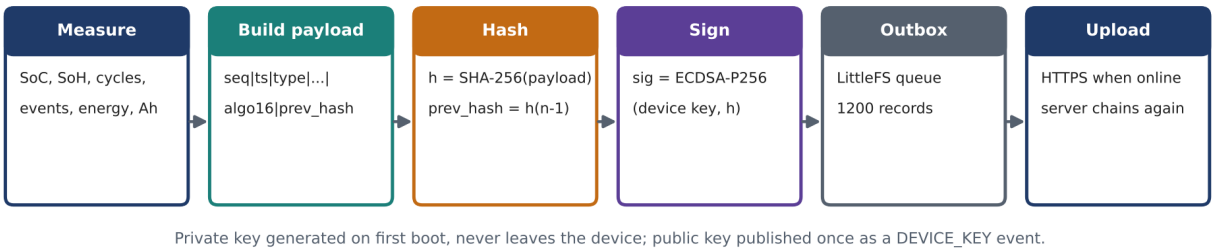


Figure 9.1 – Record pipeline on the ESP8266: measure, build payload, hash, sign, queue, upload.

9.2.1 Device identity

On first boot the ESP8266 creates its own ECDSA P-256 key pair and keeps the private key in flash; it never leaves the device. After its clock is synchronised it publishes the public key once in a `DEVICE_KEY` event that the cloud chains. The current key id is dfa8667ff212c92c (first 16 hex digits of SHA-256 of the public key).

9.2.2 Signing every record

Before a record is queued, the device builds a fixed-format payload of the values that matter, hashes and signs it:

```
seq | timestamp | record_type | event_type | cycle_count | deep | over | accident |
SoC | SoH | energy_kWh | capacity_Ah | SoH-algorithm hash (16) | previous device hash

h_n = SHA-256(payload_n)      payload_n contains h_(n−1)      sig_n = ECDSA-P256(sk, h_n)
```

- `seq` increases by one for every record the device creates, so a record that never reaches the cloud leaves a visible gap.
- Each payload contains the previous device hash, so the device keeps its own chain, independent of the server chain.
- The payload, hash, key id and 64-byte signature are stored in `raw_telemetry.dev`, which the server also hashes into its own chain.

A real record from the live system (sequence 219, 25 Oct 2026 14:12:55 UTC):

```
219|2026-10-25T14:12:55.171Z|TELEMETRY|-|224|44|0|0|99.50|96.12|555.7026|11650.427|
821963b754474cd3|27c0652a2e1a15aaa5e3f45f858c1299b3ee0fc57644949d85936b839f875228
```

9.2.3 Consequence for vehicles without connectivity

The vehicle needs no live connection. A workshop, a charging station or a phone app can carry the queued records to the cloud later; because every record is already signed and chained, the carrier cannot alter, reorder or drop records without the verification failing.

9.3 Independent Auditor Tool

`verify_device_chain.py` (Python, cryptography library) lets an auditor check the records without trusting the server or the OEM. It reads the Tier 3 response for a battery (live, with an API key or portal session, or from a saved file) and checks for every device-signed record: (1) SHA-256 of the payload equals the stored device hash; (2) the ECDSA signature verifies with the published device key; (3) each payload names the previous record's hash, and missing sequence numbers are reported as gaps; (4) the signed values equal the values stored in the database; (5) which SoH algorithm produced each value, any `SOH_ALGORITHM` changes and the cross-check results.

```
python verify_device_chain.py --bin DBP-2024HT65556-003 --api-key <TIER3_KEY>
python verify_device_chain.py --file full.json
```

9.4 Running on BMS, VCU or Telematics Only

The current build already shows that the DBP logic does not need a separate VCU: one ESP8266 runs the battery model, BMS, VCU and gateway together with no CAN bus. The next step is to make the same code portable to whichever controller a vehicle has.

Table 9.2 – Placement of the DBP core by vehicle architecture.

Vehicle has	Where the DBP core would run	Data path
BMS + VCU + telematics	VCU (as in the Raspberry Pi and ESP8266 builds)	BMS → CAN → VCU → telematics → cloud
BMS + telematics, no VCU	Telematics unit	BMS → CAN → telematics → cloud
BMS only, no telematics	BMS firmware	BMS flash → workshop download → cloud

Planned split: a **portable core** (C library) with counters, VCU metrics, events, SoH fingerprint and cross-check, device signing and hash chain and the outbox format, with no hardware calls; **thin adapters** per controller for signal sources (CAN, internal BMS variables, simulation), storage and transport (Wi-Fi, cellular, UDS download); and as **proof**, the same core built for the ESP8266 and for the STM32F103 BMS.

Chapter 10 – DBP Use Cases & Integrity Scenarios

The passport is a regulatory instrument with legal and commercial weight; its value depends on the trustworthiness of its data. This chapter analyses the main use cases and integrity scenarios and states, for each, what the implemented system does.

Table 10.1 – DBP use cases and system responses.

Use case	Actor	DBP data used	System response (as implemented)
Second-life qualification	Recycler / repurposer	SoH history, cycles, events	Tier 2 history with SoH algorithm fingerprint and cross-check results; signed records
Insurance claim	Insurer	SoC / SoH at incident time, events	Time-stamped, append-only records; ACCIDENT events chained and signed
Fleet health monitoring	Fleet operator	SoH trend, cycles, current profile	Per-BIN history via API; batch query of BINs
Regulatory audit	Authority	Annex XIII attributes, hash chain	Tier 3 view; full-chain verification from GENESIS; auditor tool for device signatures
End-of-life recovery	Recycler	BIN, chemistry, capacity, SoH	Tier 1 by QR code without login

10.1 Scenario 1 – Deliberate BMS Replacement

A workshop or owner replaces the BMS to reset accumulated SoH and cycle data or to report inflated SoH — the battery equivalent of odometer fraud.

- **Implemented:** the server rejects any record whose cycle count, throughput or event counters are lower than the previous record (HTTP 409); a replaced BMS that restarts its counters cannot write to the existing passport. A new device produces a new key, so a `DEVICE_KEY` change is visible in the chained history, and its SoH must still match the independent VCU measurement.
- **Not yet implemented:** an automatic "Data Continuity Warning" on the Tier 1 record and notification of Tier 2 stakeholders; clearing by a manufacturer-authorised Tier 3 action.

10.2 Scenario 2 – BMS Firmware Manipulation

An attacker changes the SoH calibration so that a battery at true SoH 70 % reports 90 %.

- **Implemented:** the SoH algorithm fingerprint changes and is logged at once; the independent VCU capacity measurement flags the deviation within one full cycle (SOH_CROSSCHECK_FAIL).
- **Remaining gap:** a manipulation that falsifies SoC and SoH consistently could pass the cross-check because the VCU uses the BMS SoC; an independent current sensor and voltage-based SoC in the VCU would close this. Binding the fingerprint to a signed firmware image (secure boot) is future work.

10.3 Scenario 3 – Data Injection on the CAN Bus

Forged CAN frames carry false SoH/SoC to the VCU. Mitigations: the bus is internal to the vehicle; the integrated ESP8266 path has no CAN bus at all; and an injected value that disagrees with the VCU's own measurements is caught by the cross-check. A CAN message authentication code (e.g. AUTOSAR SecOC) between BMS and VCU remains future work for Path A. On the cloud side only holders of the VCU token can write, and every device record carries the device signature.

10.4 Scenario 4 – Battery Pack Swap Between Vehicles

A pack moved to another vehicle keeps its BIN, so all later records continue the same chain and history. Recording the vehicle identity (VIN–BIN pairing) as a transfer event is supported by the event model (event type STATUS_CHANGE / MAINTENANCE) but VIN fields were not added to the record schema in this phase.

10.5 Scenario 5 – Tampering With Stored Records or Withholding Records

An insider with database access edits or deletes an old record, or a carrier drops records collected offline.

- **Edit or delete:** blocked by the append-only triggers; if bypassed by a database superuser, the portal's full-chain check and the device signatures expose the change (tested: one stored SoH value changed after signing → auditor FAIL naming the record and field).
- **Withheld records:** device sequence numbers expose the gap (tested: record 3 of 5 removed → gap 3–3 reported).

Chapter 11 – Experimental Results

11.1 Hardware Validation – TOSUN CAN SPY at ACC Chennai

The STM32F103C8T6 [21] was programmed with BMS firmware v1.0.0 implementing the same algorithms as the Simulink model. The TOSUN CAN SPY [22] decoded physical values from the bus (500 kbit/s, 120 Ω termination at both ends, twisted pair about 0.5 m, two nodes, 60 s continuous transmission).



Figure 11.1 – Bench set-up at ACC Chennai: STM32F103C8T6 BMS board and TOSUN CAN SPY.

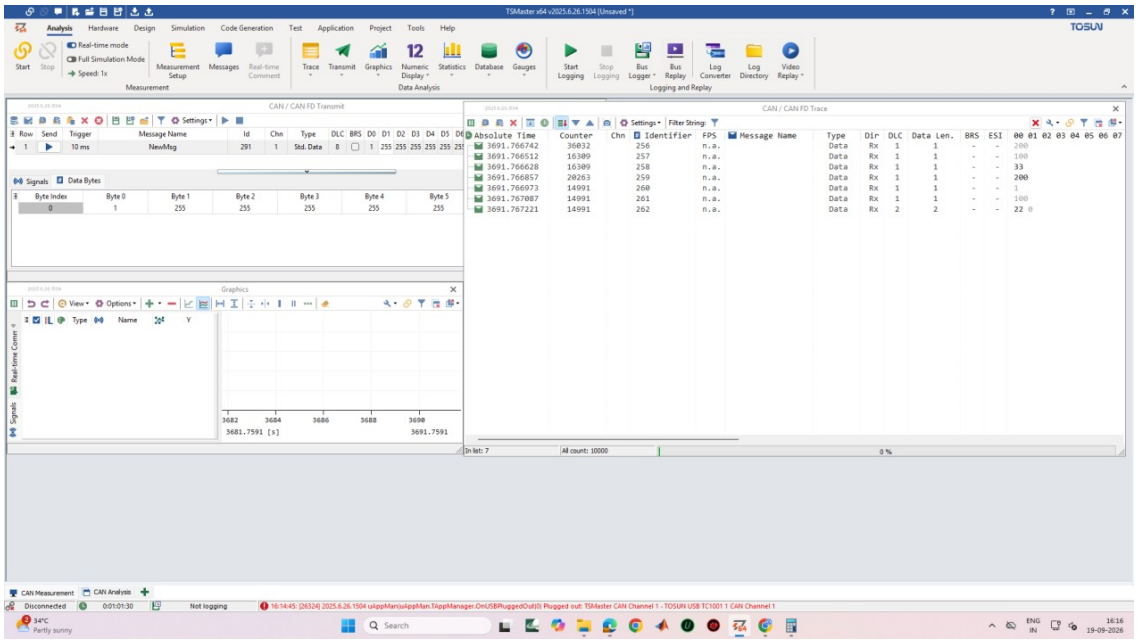


Figure 11.2 – TOSUN CAN SPY trace of the BMS frames at 500 kbit/s.

Table 11.1 – BMS values compared with the values decoded on the CAN bus (TOSUN CAN SPY at ACC Chennai).

BMS output	Value in BMS	Value on CAN	Difference
Battery voltage	200 V (static)	200 V (static)	0.0 V
Battery capacity	100 Ah (static)	100 Ah (static)	0.0 Ah
Battery current (peak)	200 A	200 A	0.0 A
SoC (end of cycle 1)	100.0 % (reset)	100.0 % (reset)	0 %
SoC (end of cycle 2)	100.0 % (reset)	100.0 % (reset)	0 %
SoH (after cycle 1)	100 %	100 %	0 %
SoH (after cycle 2)	100 %	100 %	0 %
Cycle count (end)	22	22	0
BIN	001	001	N/A

11.2 Cloud Store – Live State on 31 October 2026

Table 11.2 – Records in the live DBP store (Supabase query, 31 Oct 2026).

Battery	Source	Records	Period	Max cycles	Min SoH	Device-signed
DBP-2024HT65556-001	Real battery passport (static data + 1 record)	1	3 Oct 2026	12	98.2 %	0
DBP-2024HT65556-002	VCU simulator (Path A software)	164	Jan–May 2026 timeline	132	98.215 %	0
DBP-2024HT65556-003	ESP8266 integrated controller (Path B)	604	24–31 Oct 2026	252	95.63 %	312

Battery	Source	Records	Period	Max cycles	Min SoH	Device-signed
Total		769				312

11.3 Integrity Verification on Live Data

Table 11.3 – Integrity verification results.

Test	Data	Result
Server hash chain, full history	All 604 records of -003, portal (Tier 3), 31 Oct	PASS — chain intact from GENESIS, all fields and links
Append-only enforcement	dbp_records triggers (live query)	dbp_records_no_update (BEFORE UPDATE OR DELETE) and dbp_records_no_truncate present
Device signatures	310 device-signed records of -003 exported 31 Oct	PASS — 310 of 310 valid with key dfa8667ff212c92c
Device chain and gaps	Same 310 records, including 42 uploaded after 40 min offline	PASS — every prev link correct, 0 missing records
Signed vs stored values	Timestamp, counters, SoC, SoH, energy, capacity of 310 records	PASS — 0 differences
SoH algorithm	All TELEMETRY records	Two fingerprints: 821963b7… (278 records) and df31f211… (21 records, manipulation demo); three SOH_ALGORITHM events
Auditor tool – valid chain	5 signed records (synthetic, firmware format)	PASS
Auditor tool – value edited after signing	One stored SoH changed 99.9 → 95.0	FAIL, names the record and field
Auditor tool – record withheld	Record 3 of 5 removed	PASS with reported gap 3–3
Counter rollback	Record with lower cycle count than the previous one	Rejected with HTTP 409 by the ingest validation rule (Section 7.3)
Portal – simulated tampering	Three records of -003 altered in the /chain response (browser), 31 Oct	Detected — #300 hash mismatch; #475 link to #474 broken; #600 stored value differs (soh_pct)
Access control (live, 25 Oct)	No credentials: /public, /stakeholder, /full, /chain; forged token on /chain; POST /record without VCU token	200, 401, 403, 403; 403; 401

Figure 11.3 shows the published auditor tool (`tools/verify_device_chain.py` from the public repository, run unmodified in Python 3.12 / Pyodide with the cryptography library) on the 310 device-signed records of -003 exported from the database on 31 October 2026. Its integrity verdict is PASS; separately it reports the SoH algorithm change and the cross-check FAIL of the manipulation demonstration. On 25 October the signature

check of the 219 records then present was also repeated with the browser WebCrypto API, an implementation independent of both the firmware (BearSSL) and the auditor tool (OpenSSL).

```
$ python tools/verify_device_chain.py --file full_003.json
Battery DBP-2024HT65556-003: 310 records in window, 310 signed by the device
Device key id dfa8667ff212c92c


Device signatures: 310 valid, 0 with problems, 0 missing record(s)

SoH algorithms seen (fingerprint → records):
  DBP-SOH-EFC v1.0.0  821963b754474cd3…  278 record(s)
  DBP-SOH-EFC v1.0.0  df31f211e621c6ac…  21 record(s)
  ⚠ the SoH algorithm/calibration changed inside this window — see SOH_ALGORITHM events
  event 2026-10-24T13:05:06  SOH_ALGORITHM DBP-SOH-EFC|1.0.0|k_fade=0.0200|eol=80.0|rated_ah=60.0|offset=0.00 sha256 821963b754474cd356c27f
  event 2026-10-31T07:08:55  SOH_ALGORITHM DBP-SOH-EFC|1.0.0|k_fade=0.0200|eol=80.0|rated_ah=60.0|offset=5.00 sha256 df31f211e621c6ac58c942
  event 2026-10-31T07:09:19  SOH_CROSSCHECK_FAIL BMS 100.66% vs VCU-measured 94.06% (dev 6.60%, limit 3.0%) algo df31f211e621c6ac
  event 2026-10-31T07:33:42  SOH_ALGORITHM DBP-SOH-EFC|1.0.0|k_fade=0.0200|eol=80.0|rated_ah=60.0|offset=0.00 sha256 821963b754474cd356c27f

VCU SoH cross-check: 284 telemetry records with a result, 23 FAIL; latest: BMS 95.63 % vs VCU-measured 94.04 % (deviation 1.61 %) → PASS

VERDICT: PASS
[exit code 0 ]
```

Figure 11.3 – Independent auditor tool on the live records of -003 (31 Oct 2026): all signatures valid, no gaps, the SoH algorithm change and the cross-check FAIL reported.

11.4 SoH Cross-Check Results

Up to 31 October 2026, 284 TELEMETRY records of -003 carried a completed cross-check: 261 PASS and 23 FAIL. All 23 FAIL results belong to the deliberate manipulation in Section 11.5; no FAIL occurred during honest operation. On 24–25 October the mean absolute deviation between the BMS SoH and the VCU-measured SoH was 0.91 % and the maximum 1.43 %; on 31 October, at cycles 244–252, it was 1.55–1.61 % — inside the 3 % screening limit.

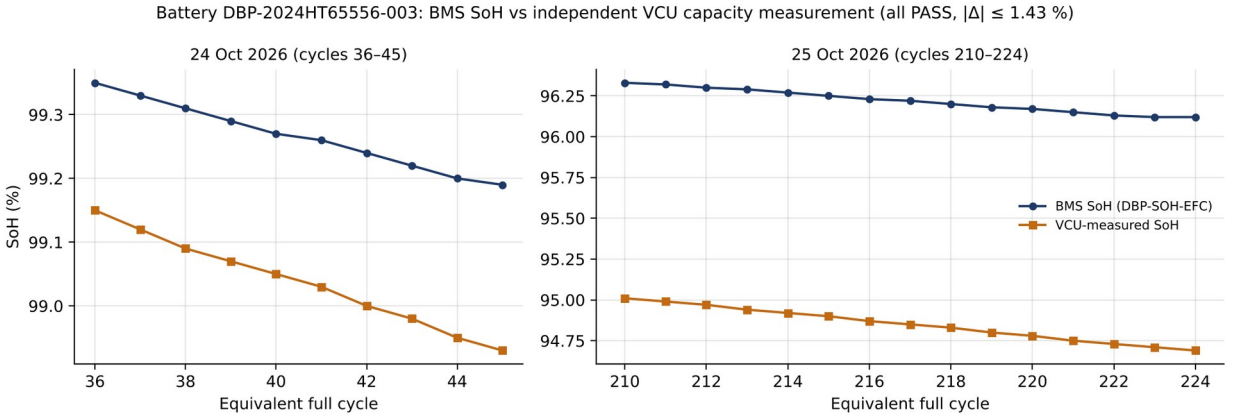


Figure 11.4 – BMS SoH versus independent VCU capacity measurement on test battery -003 (24–25 Oct 2026).

The BMS reads above the VCU measurement and the gap widens slowly: 0.20 % at cycle 36 and 1.43 % at cycle 224. Only part of this is explained by the deliberate calibration difference (cells age at 0.021 %/EFC, the BMS assumes 0.020 %/EFC, about 0.2 % after 224 cycles); the remainder is likely due to the VCU measurement method (DoD scaling of partial cycles) and should be characterised further. At the observed rate of about 0.0065 % per cycle the gap would reach the 3 % limit near cycle 460, at which point the algorithm would be flagged for recalibration — the intended behaviour of the screening check.

Cross-check results exist for cycles 36–45 (24 Oct) and 210–224 (25 Oct); the earlier records of -003 carry no cross-check result. No records were received between 24 Oct 14:28 and 25 Oct 12:58 UTC while the board was not connected; the simulated battery continued, its cumulative counters (cycles, throughput, 44 deep-discharge events) carried over, and the device sequence numbers show no missing signed record.

11.5 Live SoH Manipulation Demonstration

On 31 October 2026 the manipulation described in Section 9.1.3 was carried out on the running ESP8266 and the live cloud. Figure 11.5 shows the SoH reported by the BMS and the SoH measured independently by the VCU, as stored in the database; Table 11.4 lists the chained events.

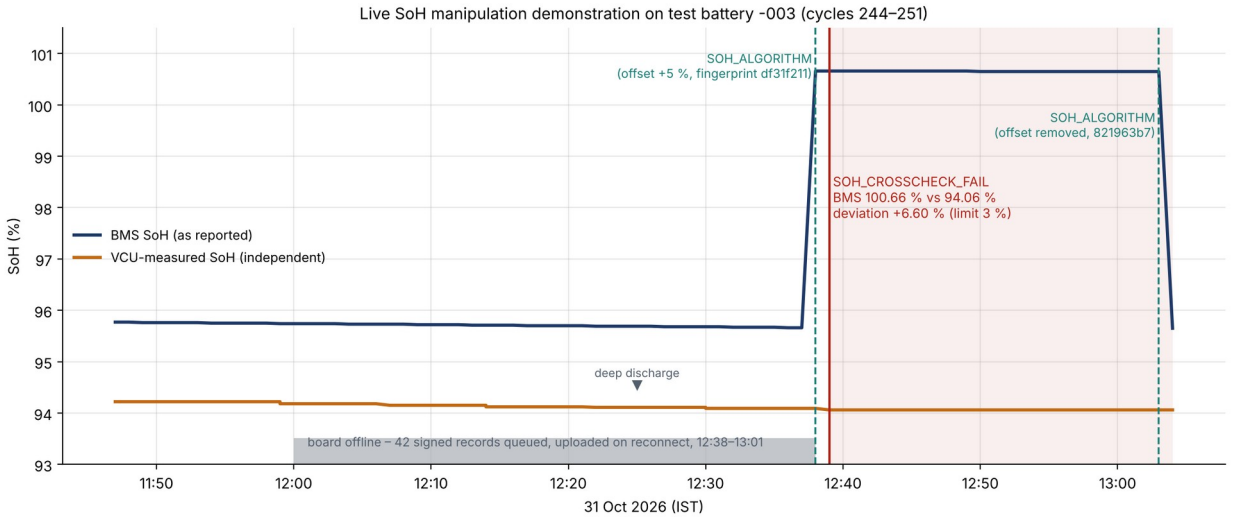


Figure 11.5 – Live demonstration on -003: +5 % SoH manipulation detected by the algorithm fingerprint at once and by the VCU cross-check at the next full cycle; restored afterwards.

Table 11.4 – Chained events of the manipulation demonstration (31 Oct 2026).

Time (UTC)	Cycle	Event (stored, hash-chained, device-signed)	Meaning
07:08:55	250	SOH_ALGORITHM … offset=5.00, sha256 df31f211…	Calibration changed: fingerprint 821963b7 → df31f211

Time (UTC)	Cycle	Event (stored, hash-chained, device-signed)	Meaning
07:09:19	251	SOH_CROSSCHECK_FAIL BMS 100.66 % vs VCU-measured 94.06 % (dev 6.60 %, limit 3.0 %)	Independent capacity measurement disagrees with the BMS
07:09–07:33	251	23 TELEMETRY records with crosscheck = FAIL	Every record during the manipulation carries the FAIL flag
07:33:42	251	SOH_ALGORITHM … offset=0.00, sha256 821963b7…	Manipulation removed; original fingerprint restored
After cycle 252	252	crosscheck = PASS, deviation 1.61 %	Normal operation resumed

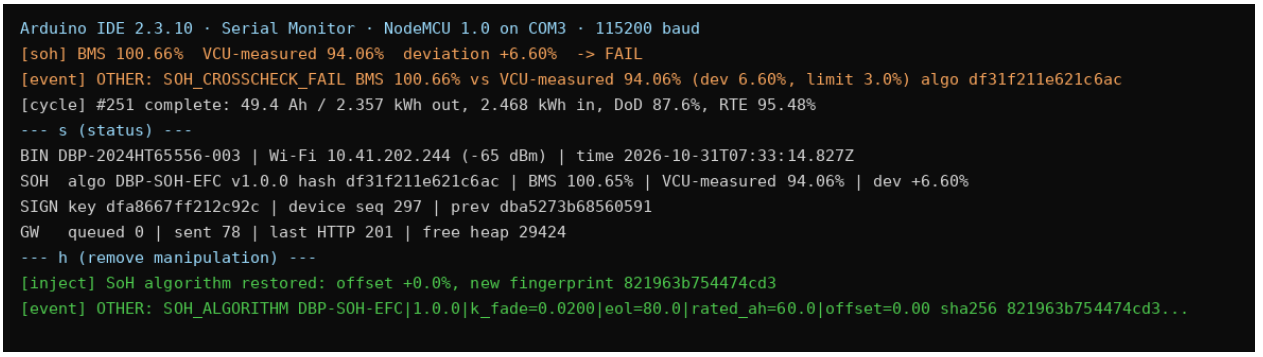


Figure 11.6 – Serial Monitor output of the ESP8266 during the demonstration (lines transcribed from the Arduino IDE).

Offline operation observed. Just before the demonstration the board had no network connection from 06:30 to 07:08 UTC. The board kept running, signed every record and held 42 of them in its flash outbox; when the connection returned they were uploaded in order within about 25 minutes, accepted by the server (HTTP 201) and verified by the auditor tool with no gaps (Figure 11.3). Earlier, one record signed on 25 October and left in the outbox was uploaded on 31 October and also verified. Both are store-and-forward cases of the kind described in Section 9.2.

Chapter 12 – Conclusion, Limitations & Future Work

12.1 Conclusion

This dissertation set out to develop a VCU-based Digital Battery Passport for tracking battery usage in accordance with EU Regulation 2023/1542 and ESPR 2027, and has delivered a working end-to-end system. The BMS controller model and STM32 firmware produce the seven DBP signals, validated on the CAN bus at ACC Chennai. Two VCU implementations compute the Annex VII in-use parameters and deliver them to a cloud service that validates every record, links it into a per-battery SHA-256 hash chain and stores it in an append-only database. A public portal presents the passport in three access tiers and verifies the complete chain from GENESIS in the user's own browser.

Beyond storing data, the work addressed trusting it. The system addresses SoH honesty and offline operation with mechanisms that run on the vehicle: every SoH value names the algorithm and calibration that produced it, an independent capacity measurement checks the BMS SoH every full cycle, and every record is signed and chained on the device so that it can wait offline and still be verified by an auditor who trusts neither the cloud operator nor the OEM. On live data all 604 server records of the test battery and all 310 device signatures verified; a deliberately inflated SoH (+5 %) was flagged by the algorithm fingerprint at once and by the independent cross-check at the next full cycle (deviation 6.60 %), and 42 records signed during a 40-minute loss of connectivity arrived intact. Controller independence is shown in principle by the single-MCU build and will be completed by the planned portable core.

The design choices are deliberately open and inexpensive — a public repository, free-tier hosting and a sub-USD 5 microcontroller — in line with the project's goal of a passport approach that small pack assemblers and integrators, not only large OEMs, can adopt.

12.2 Limitations

- All battery data on the ESP8266 path is simulated (13S, 48 V, 60 Ah model at 60× speed). The static passport of test battery -003 was aligned with this model on 31 Oct 2026 (48 V, 60 Ah, 2.88 kWh, LMT); that the update was possible at all shows that static data is not yet write-once.

- The cross-check uses the BMS SoC; an OEM falsifying SoC and SoH consistently could pass it.
- The SoH fingerprint shows what the device says it runs; it is not yet bound to the firmware image by secure boot.
- The device private key is in ordinary flash; production needs a secure element or protected MCU storage.
- Without a real-time clock the ESP8266 cannot timestamp records after a reboot until it reaches a time server.
- Tier 2/3 use prototype accounts and HMAC session tokens instead of OAuth 2.0 and X.509; the server does not yet verify device signatures on ingest.
- Free-tier hosting sleeps (Render) and pauses (Supabase) after inactivity; acceptable for a demonstrator, not for production.
- The ESP8266 is single-core; signing takes up to about 1 s per record — sufficient for one record per minute.

12.3 Future Work

Table 12.1 – Future work programme.

Area	Task	Platform
SoH trust	Independent current sensor and voltage-based SoC in the VCU; bind the SoH fingerprint to a signed firmware hash	VCU / BMS firmware
Offline	Workshop download of queued records over UDS; DS3231 RTC for offline timestamps	ESP8266 / STM32
Portability	Split the DBP core into a portable C library and build it for the STM32 BMS	C library
Cloud	Verify device signatures on ingest; database trigger to make battery_passport write-once; SoH-trend lifecycle staging; continuity warnings	API / PostgreSQL
Identity	OAuth 2.0 for Tier 2, X.509 client certificates for Tier 3; device keys in a secure element	API / hardware
Standards	Map fields to the EU DPP data model and IEC 63386 when published; EPCIS-conformant BIN	Data model
Validation	End-to-end bench test BMS → CAN → Raspberry Pi VCU → cloud with real cells; CAN bus-off fault injection; temperature characterisation	ACC Chennai lab
Dissemination	Publish a white paper and keep the reference code in the public repository	—

12.4 Summary of Contributions

1. A MATLAB Simulink BMS and VCU model producing the DBP signals, with STM32F103C8T6 firmware validated on the CAN bus at 500 kbit/s.
2. A 60-requirement DBP specification (StRS, SyRS, CRS, SyRS-CLD and new trust requirements SyRS-T) using INCOSE methodology, with traceable verification.
3. Three documented SE decisions: cloud-side lifecycle computation, hash chain plus signatures instead of blockchain, and platform selection.
4. A live, open cloud DBP service with canonical-JSON SHA-256 chaining, database-enforced append-only storage, server-enforced access tiers and full-chain verification in the browser.
5. An ESP8266 integrated BMS + VCU + gateway with an SoH algorithm fingerprint, an independent SoH cross-check, ECDSA-signed device-chained records and an offline outbox, plus an independent auditor tool — a concrete answer to the questions of how to trust SoH and how to keep a passport without permanent connectivity.

Appendix A – Demonstration Procedure

About 25 minutes, mostly waiting for two simulated cycles; start it in advance.

1. **Board running.** Arduino IDE → Serial Monitor (115200 baud) → type `s`; point out the SOH line (algorithm, hash, BMS vs VCU-measured SoH) and the SIGN line (key id, device sequence).
2. **Published key.** Type k; the same key appears in the DEVICE_KEY event in the portal's Tier 3 history.
3. **Honest cross-check.** Wait for `[soh] … -> PASS` after a full cycle (about 8 minutes).
4. **Manipulate the BMS.** Type h; a new fingerprint and a SOH_ALGORITHM event appear at once.
5. **Caught.** After the next full cycle the Serial Monitor shows `-> FAIL` and a SOH_CROSSCHECK_FAIL event reaches the cloud.
6. **Audit.** Run verify_device_chain.py: all signatures valid, two algorithm fingerprints in the window, cross-check FAIL listed.
7. **Tamper evidence.** Edit one SoH value in a saved full.json and run the verifier again: it names the altered record.
8. **Portal.** Log in as Regulator, look up -003: chain intact from GENESIS.
9. **Restore.** Type h again to remove the manipulation.

Appendix B – Software Artefacts

Table 12.2 – Location of the project software.

Artefact	Location
DBP API (main.py), portal (docs/index.html), VCU simulator, database migration	github.com/snithyamb/dbp-cloud
Live API / portal	dbp-cloud.onrender.com · snithyamb.github.io/dbp-cloud
ESP8266 firmware v2 (dbp_esp8266.ino)	Documents\Arduino\dbp_esp8266\ (secrets.h kept local, never committed)
Auditor tool (verify_device_chain.py)	Documents\Arduino\dbp_tools\
Simulink models (New_BMS_2.slx, New_VCU.slx)	MATLAB package (Simulink BMS and VCU models)

Appendix C – Particulars of the Supervisor and Additional Examiner

Table 12.3 – Particulars of the supervisor and additional examiner.

	Supervisor	Additional Examiner
Name	Saravana Kumar	Ravikumar M
Qualification	M.Tech – Aerospace Engineering	B.E – Automobile Engineering
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E-mail	Lsaravanakumar4791@gmail.com	ravi.irtt@gmail.com

Signature of Student: _____ Signature of Supervisor: _____

Signature of Additional Examiner: _____

Nithyanandham S Saravana Kumar Ravikumar M

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Glossary

Technical terms and abbreviations used in this report, with the pages on which they first occur.

Term	Meaning	Page(s)
Annex XIII	Annex of Regulation (EU) 2023/1542 listing the information held in the battery passport	3, 5, 6
API	Application Programming Interface	2, 6, 7
Append-only	Storage in which records can be added but never changed or deleted; enforced here by database triggers	2, 5, 9
Auditor tool	Published script (verify_device_chain.py) that checks device signatures, chain links and gaps without trusting the server	3, 10, 13
BIN	Battery Identification Number	9, 12, 15
BMS	Battery Management System	2, 3, 5
CAN	Controller Area Network (ISO 11898)	2, 3, 5
Canonical JSON	Fixed serialisation of a record (sorted keys, fixed number format) so that the same data always gives the same hash	9, 23, 24
CORS	Cross-Origin Resource Sharing	21, 31
Cross-check (SoH)	Comparison of the BMS SoH with the capacity measured independently by the VCU over a full cycle	33, 35, 36
DBP	Digital Battery Passport	1, 2, 5
DoD	Depth of Discharge	33, 36, 46
ECDSA	Elliptic Curve Digital Signature Algorithm (here on curve P-256)	8, 13, 21
EFC	Equivalent Full Cycle	16, 33, 36
ESPR	Ecodesign for Sustainable Products Regulation	5, 11, 24
EU	European Union	1, 2, 3
Fingerprint (SoH algorithm)	SHA-256 hash of the SoH algorithm name, version and calibration, carried in every record	13, 35, 36
GENESIS	Fixed prev_hash value of the first record of each battery chain	13, 25, 30
Hash chain	Sequence of records in which each record contains the hash of the previous one, so that any change breaks the chain	2, 5, 6
HMAC	Hash-based Message Authentication Code	26, 27, 34
HTTPS	HTTP over Transport Layer Security	9, 21, 32
INCOSE	International Council on Systems Engineering	2, 3, 11
JSON	JavaScript Object Notation	9, 31, 50
LMT	Light Means of Transport	5, 48
LUT	Look-Up Table	15, 18
MCU	Microcontroller Unit	2, 9, 13
OCV	Open-Circuit Voltage	33
Outbox	Flash-memory queue on the ESP8266 that holds signed records until they can be uploaded	33, 35, 39
P-256	NIST elliptic curve secp256r1	8, 13, 35
RTC	Real-Time Clock	49

Term	Meaning	Page(s)
SE	Systems Engineering	2, 8, 11
SHA-256	Secure Hash Algorithm with a 256-bit output (FIPS 180-4)	2, 5, 8
SoC	State of Charge	9, 11, 12
SoCE	State of Certified Energy	6, 16, 33
SoH	State of Health	2, 3, 5
Store and forward	Keeping records on the vehicle while offline and uploading them later in order	33
StRS / SyRS	Stakeholder / System Requirements Specification	11, 50
Tier 1 / 2 / 3	Passport access levels: public; persons with a legitimate interest; authorities and notified bodies	6, 9, 19
TLS	Transport Layer Security	13, 34
UDS	Unified Diagnostic Services (ISO 14229)	35, 39, 49
VCU	Vehicle Control Unit	1, 2, 3

Checklist of Items for the Final Dissertation / Project / Project Work Report

This checklist is to be attached as the last page of the final report.

This checklist is to be duly completed, verified and signed by the student.

S. No.	Item	Yes / No
1	Is the final report neatly formatted with all the elements required for a technical Report?	Yes
2	Is the Cover page in proper format as given in Annexure A?	Yes
3	Is the Title page (Inner cover page) in proper format?	Yes
4	(a) Is the Certificate from the Supervisor in proper format? (b) Has it been signed by the Supervisor?	Yes Yes
5	Is the Abstract included in the report properly written within one page? Have the technical keywords been specified properly?	Yes Yes
6	Is the title of your report appropriate? The title should be adequately descriptive, precise and must reflect scope of the actual work done. Uncommon abbreviations / Acronyms should not be used in the title	Yes
7	Have you included the List of abbreviations / Acronyms?	Yes
8	Does the Report contain a summary of the literature survey?	Yes
9	Does the Table of Contents include page numbers? Are the Pages numbered properly? (Ch. 1 should start on Page # 1) Are the Figures numbered properly? (Figure Numbers and Figure Titles should be at the bottom of the figures) Are the Tables numbered properly? (Table Numbers and Table Titles should be at the top of the tables) Are the Captions for the Figures and Tables proper? Are the Appendices numbered properly? Are their titles appropriate	Yes Yes Yes Yes Yes Yes
10	Is the conclusion of the Report based on discussion of the work?	Yes
11	Are References or Bibliography given at the end of the Report? Have the References been cited properly inside the text of the Report? Are all the references cited in the body of the report	Yes Yes Yes
12	Is the report format and content according to the guidelines? The report should not be a mere printout of a PowerPoint Presentation, or a user manual. Source code of software need not be included in the report.	Yes

Declaration by Student:

I certify that I have properly verified all the items in this checklist and ensure that the report is in proper format as specified in the course handout.

Place: Chennai

Date: _____

Signature of the Student: _____

Name: Nithyanandham S

ID No.: 2024HT65556